

Industrial Control System (ICS) Network – Report

CSCE 4230 – Introduction to Cybersecurity (Spring 2026)

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1. Philosophy of the Design

This network was designed around one core idea: a compromised IT zone must never mean a compromised OT zone.

In industrial environments, the Operational Technology (OT) network controls physical processes, which are machines, sensors, and controllers. If an attacker reaches OT, the damage is not just digital; it can be physical and irreversible. Because of this, the entire design was built on the principle of Defense in Depth (multiple independent layers of security), so that breaking through one layer does not give access to everything.

The five guiding principles of this design are:

1. Zero Trust Between Zones

No zone trusts another by default. Every connection between zones is explicitly permitted or denied by ACLs and firewall rules. Even the Engineering zone, which needs limited OT access, cannot reach OT freely since it must go through the Jump Server.

2. OT Air-Gap by Policy

The OT zone (Zone C) is treated as if it were physically isolated. It has no internet access under any circumstance. It cannot initiate connections to IT or ENG zones. The only inbound path to OT is SSH through the Jump Server in the DMZ.

3. DMZ as the Controlled Gateway

DMZ (Zone D) acts as a buffer between the internal zones and the outside world. All critical shared services such as logging, software updates, and the Jump Server stay there. Nothing from the internet reaches internal zones directly.

4. Centralized Authentication and Logging

All devices authenticate through a central AAA server using TACACS+. All logs from all devices go to a central Logging Server in the DMZ. This means every login, every command, and every dropped packet is recorded in one place.

5. Least Privilege

Each zone can only reach what it absolutely needs. IT can reach DMZ and ENG but not OT. ENG can reach OT only via Jump Server. The remote site can only communicate with the main site through an encrypted IPsec VPN tunnel.

2. Project Objectives

The main objective of this project is to design and implement a secure enterprise and industrial network infrastructure using Cisco Packet Tracer. The project applies cybersecurity concepts practically through network segmentation, traffic filtering, secure management, and controlled communication between zones.

The project objectives include:

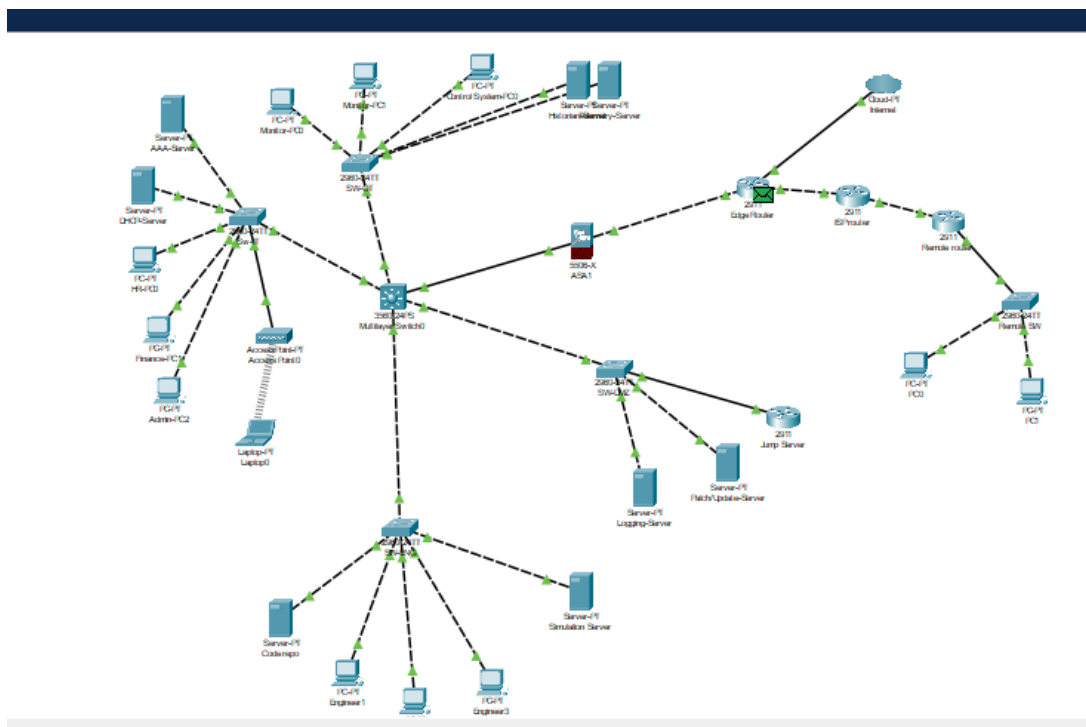
- Designing a segmented network topology using VLANs
- Implementing Inter-VLAN routing using a multilayer switch
- Separating IT, Engineering, OT, and DMZ environments
- Applying Access Control Lists (ACLs) to restrict unauthorized communication
- Configuring firewall protection using Cisco ASA 5506-X
- Preventing direct internet access to the OT network
- Implementing secure remote management using SSH
- Configuring centralized authentication using AAA/TACACS+
- Designing a secure site-to-site VPN infrastructure
- Applying cybersecurity principles such as Defense in Depth and Least Privilege
- Testing network connectivity and security restrictions between zones

3. Network Design and Topology

The network was designed using segmented architecture to improve security, manageability, and traffic isolation. The infrastructure consists of multiple zones, each serving a different operational purpose. VLAN segmentation was used to logically separate the network into isolated broadcast domains, while Inter-VLAN routing enabled controlled communication between authorized networks.

Working Demo

(<https://drive.google.com/drive/folders/14N2T0cv0d3z3Ox14SgT3hg8EhvZ87Tb5?usp=sharing>)



Core-SW

PhysicalConfigCLIAttributes

IOS Command Line Interface

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
Switch(config)#interface range fa0/1 - 5
Switch(config-if-range)#switchport mode trunk

Switch(config-if-range)#switchport trunk native vlan 99
Switch(config-if-range)#switchport trunk allowed vlan 10,20,30,40,99
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up

Switch(config-if-range)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/1 (99),
with Switch FastEthernet0/6 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/3 (99),
```

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Figure 1: Full Network Topology

3.1 Zone Architecture

The network contains four primary zones in addition to a remote site connection:

Zone	Name	VLAN	Subnet	Devices
Zone A	IT Zone	10	192.168.10.0/24	HR-PC0, Finance-PC1, Admin-PC2, DHCP-Server, AAA-Server, Access Point, Laptop
Zone B	Engineering Zone	20	192.168.20.0/24	Engineer1, Engineer2, Engineer3, Code Repo, Simulation Server
Zone C	OT Zone	30	192.168.30.0/24	Monitor-PC0, Monitor-PC1, Control-System-PC0, Historian-Server, Telemetry-Server
Zone D	DMZ	40	192.168.40.0/24	Logging-Server, Jump Server, Patch/Update-Server
Remote Site	Remote	—	192.168.50.0/24	PC0, PC1

Table 1: Network Zone Summary

3.2 Core Infrastructure Devices

The following core devices form the backbone of the network:

Device	Function
3560 Multilayer Switch	Core switch which performs VLAN routing and switching via SVIs
2911 Edge Router	Connects the network to external networks; hosts VPN services. A router was used instead of Cloud-PT for reliable IPsec tunnels and ping between sites.
ASA 5506-X Firewall	Provides firewall filtering and security policies between zones
AAA Server	Centralized authentication via TACACS+ (credentials: admin / Cisco123)
Logging Server	Centralized event and log collection (192.168.40.30)
Jump Server	Secure SSH gateway for accessing OT systems from ENG zone (replaced server with router for SSH capability)
Remote Router	VPN peer for remote site (192.168.50.0/24) over IPsec

Table 2: Core Infrastructure Devices

Note on Router vs. Cloud-PT internet: During implementation, Cisco Packet Tracer's Cloud-PT was found to be unreliable for IPsec. That means packets were not routed and tunnels could not be established. The edge router (2911) was substituted as an internet gateway, providing consistent routing, working IPsec tunnels, and reliable ping results between sites.

4. VLAN Segmentation

Virtual Local Area Networks (VLANs) were implemented to logically separate the network into multiple isolated broadcast domains. VLAN segmentation improves network performance, security, and manageability by preventing unnecessary broadcast traffic from spreading across the entire infrastructure.

Each zone is assigned a dedicated VLAN:

- VLAN 10 for IT Zone (Zone A): 192.168.10.0/24
- VLAN 20 for Engineering Zone (Zone B): 192.168.20.0/24
- VLAN 30 for OT Zone (Zone C): 192.168.30.0/24
- VLAN 40 for DMZ (Zone D): 192.168.40.0/24
- VLAN 99 for Native and Management VLAN

Access ports were assigned to their appropriate VLANs depending on the connected devices. Trunk links were configured between switches to allow multiple VLANs to pass across the infrastructure.

```
Switch#show interfaces trunk
Port      Mode      Encapsulation  Status        Native vlan
Fa0/1     on        802.1q         trunking      99
Fa0/2     on        802.1q         trunking      99
Fa0/3     on        802.1q         trunking      99
Fa0/4     on        802.1q         trunking      99
Fa0/5     on        802.1q         trunking      99

Port      Vlans allowed on trunk
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,30,40,99
Fa0/4     10,20,30,40,99
Fa0/5     10,20,30,40,99

Port      Vlans allowed and active in management domain
Fa0/1     10,20,30,40,99
Fa0/2     10,20,30,40,99
Fa0/3     10,20,30,40,99
Fa0/4     10,20,30,40,99
Fa0/5     10,20,30,40,99

Port      Vlans in spanning tree forwarding state and not pruned
--More--
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/2 (99),
with Switch FastEthernet0/4 (10).
Fa0/1     10,20,30,40
Fa0/2     10,20,30,40,99
Fa0/3     10,20,30,40
Fa0/4     10,20,30,40
Fa0/5     10,20,30,40,99

Switch#
Switch#
Switch#
```

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Figure 2: VLAN Configuration on Core Switch

```
Switch>en
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (1), with
Switch FastEthernet0/4 (99).

Switch(config)#int fa0/4
Switch(config-if)#switchport mode trunk
Switch(config-if)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/4 (1), with
Switch FastEthernet0/4 (99).

Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/4 on
VLAN0099. Port consistency restored.

%SPANTREE-2-UNBLOCK_CONSIST_PORT: Unblocking FastEthernet0/4 on VLAN0001. Port
consistency restored.

Switch(config-if)#switchport trunk allowed vlan 10,20,30,40,99
Switch(config-if)#
```

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Figure 3: Trunk Link Configuration

4.1 Switch Port Assignments

Trunk port assignments on the Core Switch (3560):

- SW-IT: connected on FastEthernet 0/2
- SW-ENG: connected on FastEthernet 0/1
- SW-DMZ: connected on FastEthernet 0/3
- SW-OT: connected on FastEthernet 0/4
- ASA Firewall: connected on the trunk port at 0/5 (between router and switch)

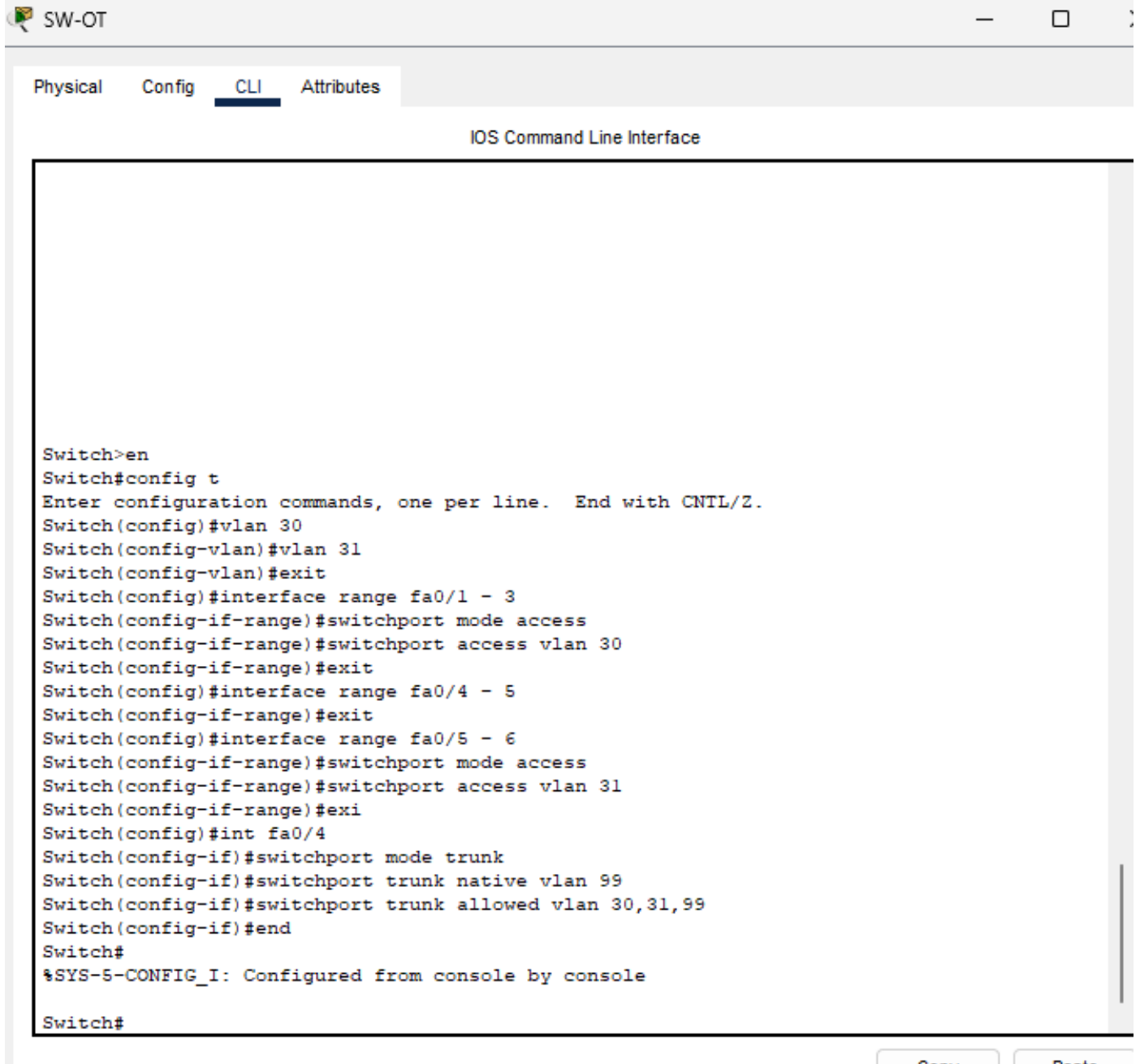


Figure 4: SW-IT Trunk Configuration

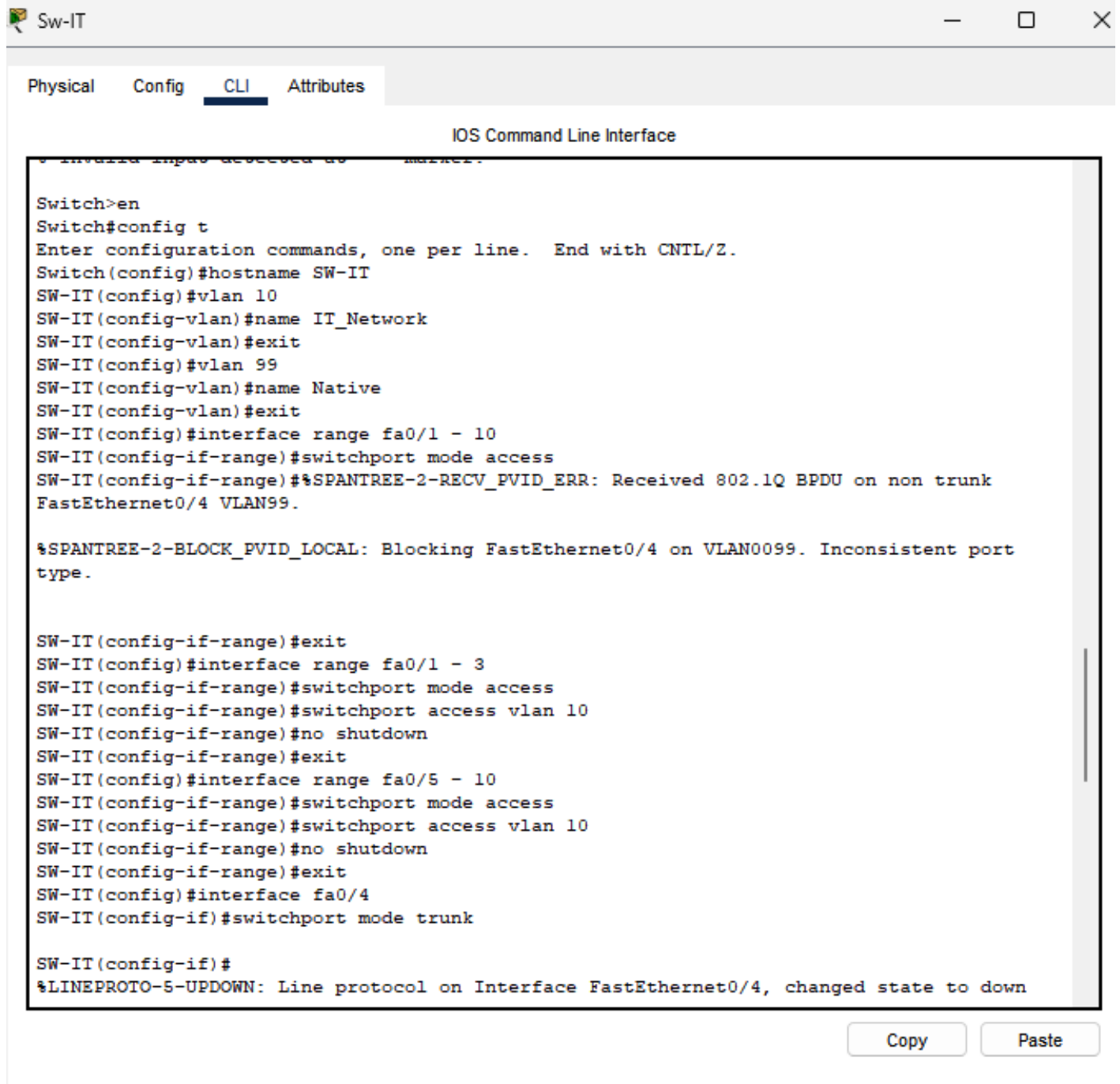


Figure 5: SW-IT Port Detail

5. Inter-VLAN Routing

Inter-VLAN routing was implemented using the Cisco 3560 multilayer switch. Switched Virtual Interfaces (SVIs) were configured for each VLAN to provide gateway functionality and enable controlled communication between authorized VLANs.

Each VLAN interface was assigned an IP address to act as the default gateway for devices within that subnet:

- VLAN 10 SVI: 192.168.10.1 (IT Zone gateway)
- VLAN 20 SVI: 192.168.20.1 (Engineering Zone gateway)
- VLAN 30 SVI: 192.168.30.1 (OT Zone gateway)
- VLAN 40 SVI: 192.168.40.1 (DMZ gateway)

Spanning Tree Protocol (STP) was also configured to prevent broadcast storms from redundant switch paths. STP automatically detects all paths between switches, blocks redundant links, and unblocks backup paths upon link failure.

6. Access Control Lists (ACLs)

ACLs were configured on both the Core Switch and the Edge Router to filter traffic and enforce communication restrictions between network zones. ACLs improve security by permitting or denying packets based on source addresses, destination addresses, and protocols.

The implemented ACL policies include:

- Blocking direct communication from the IT zone (Zone A) to the OT zone (Zone C)
- Restricting internet access from the OT network entirely
- Limiting Engineering zone access to OT resources. That means all OT access must go through the Jump Server via SSH
- Controlling communication between internal zones and the DMZ

```

Switch#show ip access-lists
Extended IP access list OT_RESTRICT
 10 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 20 deny ip 192.168.30.0 0.0.0.255 192.168.11.0 0.0.0.255
 30 deny ip 192.168.30.0 0.0.0.255 192.168.12.0 0.0.0.255
 40 deny ip 192.168.30.0 0.0.0.255 192.168.20.0 0.0.0.255
 50 deny ip 192.168.30.0 0.0.0.255 192.168.21.0 0.0.0.255
 60 deny ip 192.168.31.0 0.0.0.255 192.168.10.0 0.0.0.255
 70 deny ip 192.168.31.0 0.0.0.255 192.168.11.0 0.0.0.255
 80 deny ip 192.168.31.0 0.0.0.255 192.168.12.0 0.0.0.255
 90 deny ip 192.168.31.0 0.0.0.255 192.168.20.0 0.0.0.255
100 deny ip 192.168.31.0 0.0.0.255 192.168.21.0 0.0.0.255
110 permit ip any any
Extended IP access list ENG_TO_OT
 10 permit ip 192.168.20.0 0.0.0.255 host 192.168.40.10
 20 permit ip 192.168.21.0 0.0.0.255 host 192.168.40.10
 30 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 40 deny ip 192.168.20.0 0.0.0.255 192.168.31.0 0.0.0.255
 50 deny ip 192.168.21.0 0.0.0.255 192.168.30.0 0.0.0.255
 60 deny ip 192.168.21.0 0.0.0.255 192.168.31.0 0.0.0.255
 70 permit ip any any
Extended IP access list DMZ_TO_OT
 10 permit ip host 192.168.40.10 192.168.30.0 0.0.0.255
 20 permit ip host 192.168.40.10 192.168.31.0 0.0.0.255
 30 deny ip 192.168.40.0 0.0.0.255 192.168.30.0 0.0.0.255
 40 deny ip 192.168.40.0 0.0.0.255 192.168.31.0 0.0.0.255
 50 permit ip any any

Switch#

```

Figure 6: ACL Verification — show ip access-lists on Core Switch and Router

6.1 ACL Test Results

Test 1: IT restrict (IT cannot reach OT):

- ping 192.168.30.1 (OT zone) from HR-PC0 → BLOCKED
- ping 192.168.20.1 (ENG zone) from HR-PC0 → ALLOWED
- ping 192.168.40.1 (DMZ zone) from HR-PC0 → ALLOWED

```
-----
CoreRouter#sh ip access-lists
Extended IP access list OUTBOUND
 10 deny ip 192.168.30.0 0.0.0.255 any (8 match(es))
 20 deny ip 192.168.31.0 0.0.0.255 any
 30 permit ip 192.168.10.0 0.0.0.255 any (12 match(es))
 40 permit ip 192.168.11.0 0.0.0.255 any
 50 permit ip 192.168.12.0 0.0.0.255 any
 60 permit ip 192.168.20.0 0.0.0.255 any
 70 permit ip 192.168.21.0 0.0.0.255 any
 80 permit ip 192.168.40.0 0.0.0.255 any
Extended IP access list INBOUND
 10 permit tcp any any established
 20 permit icmp any any echo-reply
 30 permit udp any any eq isakmp
 40 permit esp any any
 50 deny ip any any
```

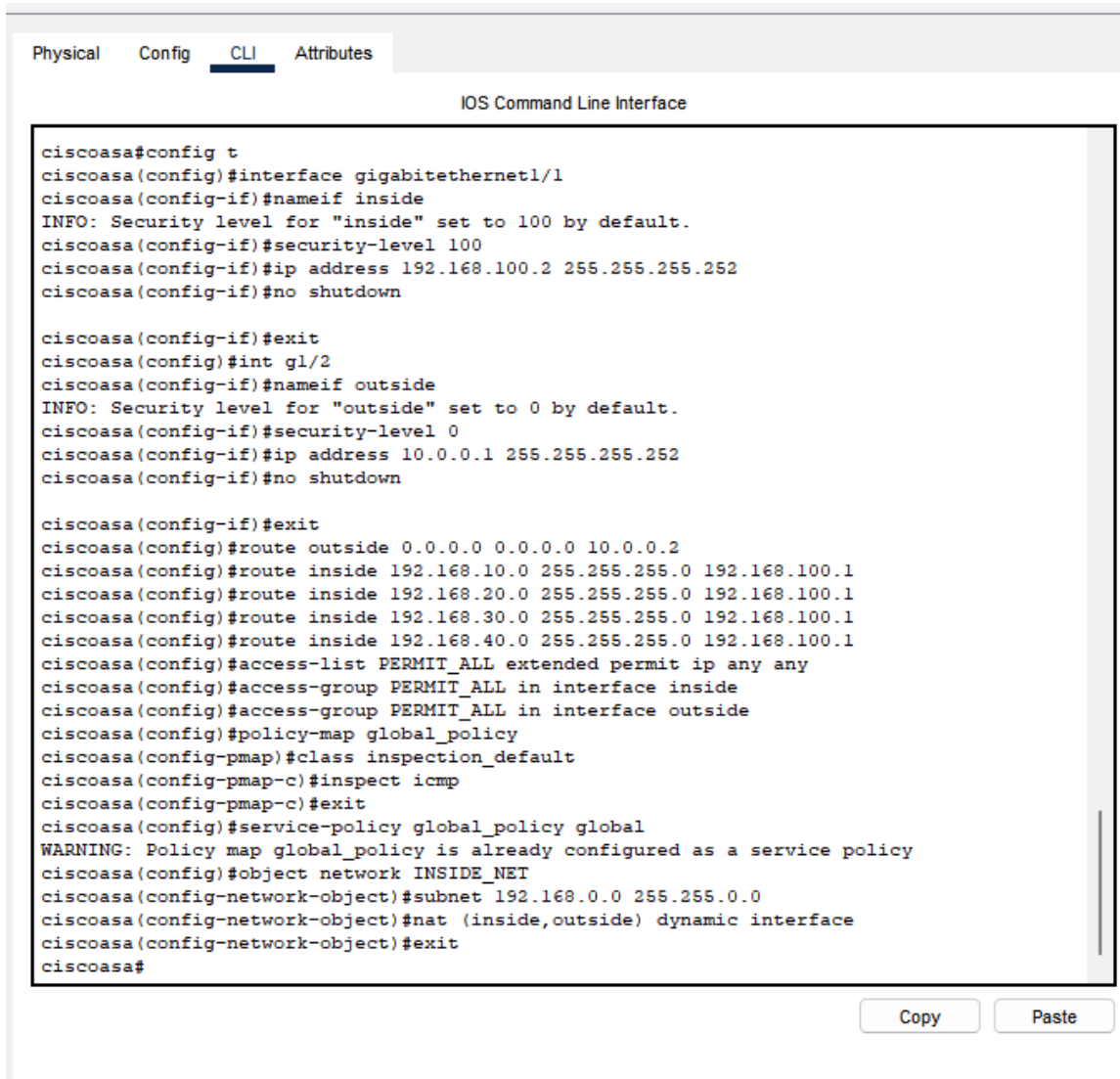
Figure 7: Router ACLs — Access Lists Confirmed Active

7. ASA Firewall Configuration

The Cisco ASA 5506-X firewall was implemented to provide perimeter security and traffic filtering between internal networks and external connections. The ASA firewall was configured with inside and outside interfaces, routing policies, and packet inspection services.

The firewall configuration included:

- Interface security levels (inside, outside, DMZ)
- NAT configuration for outbound traffic
- ICMP inspection
- Access-group policies enforcing zone isolation
- Static routing for internal zone reachability



```
ciscoasa#config t
ciscoasa(config)#interface gigabitethernet1/1
ciscoasa(config-if)#nameif inside
INFO: Security level for "inside" set to 100 by default.
ciscoasa(config-if)#security-level 100
ciscoasa(config-if)#ip address 192.168.100.2 255.255.255.252
ciscoasa(config-if)#no shutdown

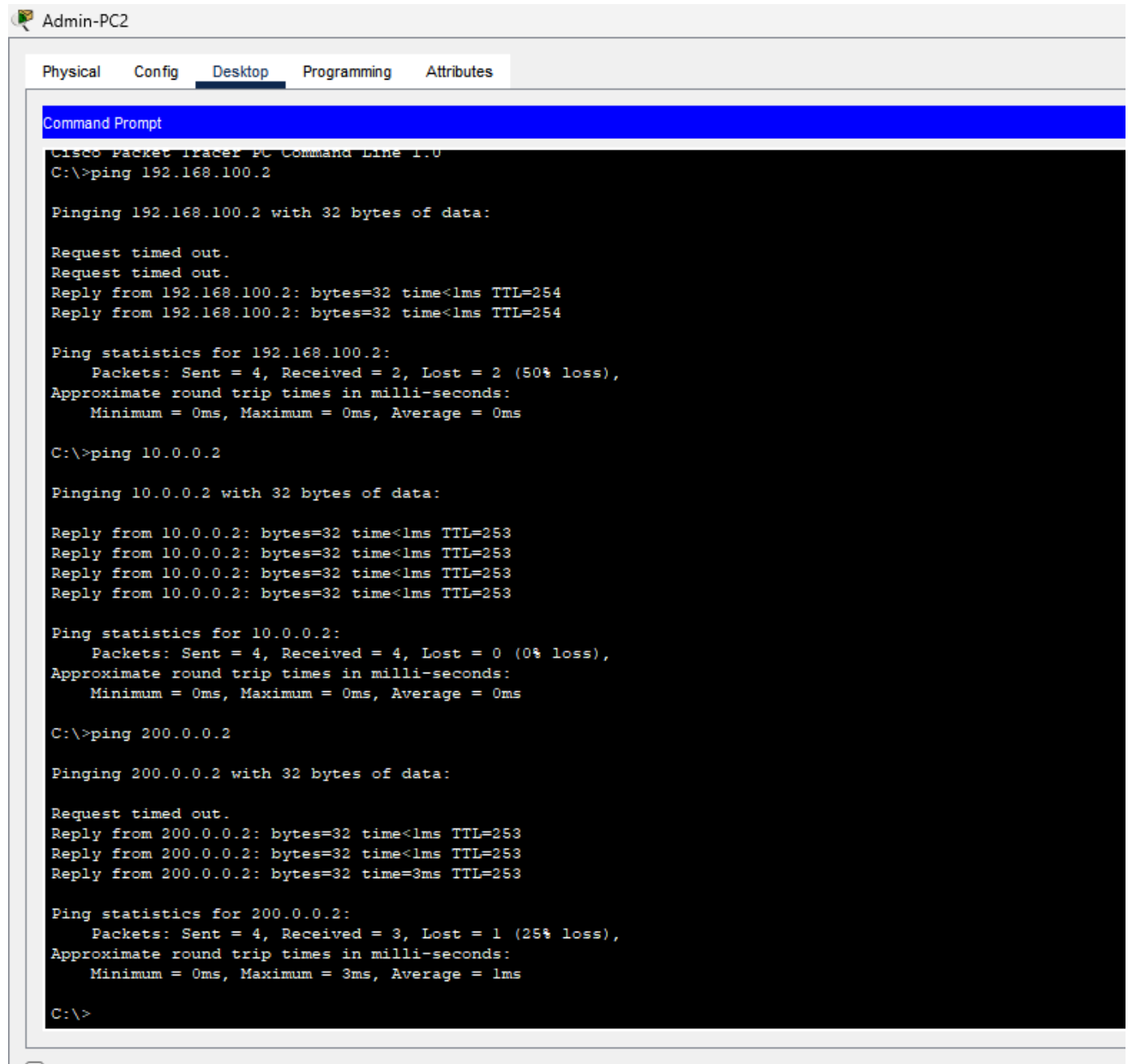
ciscoasa(config-if)#exit
ciscoasa(config)#int gi/2
ciscoasa(config-if)#nameif outside
INFO: Security level for "outside" set to 0 by default.
ciscoasa(config-if)#security-level 0
ciscoasa(config-if)#ip address 10.0.0.1 255.255.255.252
ciscoasa(config-if)#no shutdown

ciscoasa(config-if)#exit
ciscoasa(config)#route outside 0.0.0.0 0.0.0.0 10.0.0.2
ciscoasa(config)#route inside 192.168.10.0 255.255.255.0 192.168.100.1
ciscoasa(config)#route inside 192.168.20.0 255.255.255.0 192.168.100.1
ciscoasa(config)#route inside 192.168.30.0 255.255.255.0 192.168.100.1
ciscoasa(config)#route inside 192.168.40.0 255.255.255.0 192.168.100.1
ciscoasa(config)#access-list PERMIT_ALL extended permit ip any any
ciscoasa(config)#access-group PERMIT_ALL in interface inside
ciscoasa(config)#access-group PERMIT_ALL in interface outside
ciscoasa(config)#policy-map global_policy
ciscoasa(config-pmap)#class inspection_default
ciscoasa(config-pmap-c)#inspect icmp
ciscoasa(config-pmap-c)#exit
ciscoasa(config)#service-policy global_policy global
WARNING: Policy map global_policy is already configured as a service policy
ciscoasa(config)#object network INSIDE_NET
ciscoasa(config-network-object)#subnet 192.168.0.0 255.255.0.0
ciscoasa(config-network-object)#nat (inside,outside) dynamic interface
ciscoasa(config-network-object)#exit
ciscoasa#
```

Figure 8: ASA 5506-X Firewall Configuration

7.1 Zone-Based Firewall (ZBF)

In addition to the ASA, Zone-Based Firewall policies were also configured on the Edge Router to enforce traffic control at the perimeter. The ZBF policies complement the ASA by providing an additional layer of traffic inspection and control.



The screenshot shows a Cisco Packet Tracer PC Command Line interface for a device named 'Admin-PC2'. The interface has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The command prompt shows the following sequence of commands and outputs:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.100.2: bytes=32 time<1ms TTL=254
Reply from 192.168.100.2: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 200.0.0.2: bytes=32 time<1ms TTL=253
Reply from 200.0.0.2: bytes=32 time<1ms TTL=253
Reply from 200.0.0.2: bytes=32 time=3ms TTL=253

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

Figure 9: Zone-Based Firewall (ZBF) Testing

8. SSH Secure Management

Secure Shell (SSH) was configured to provide encrypted remote management access to network devices. SSH was selected over Telnet because Telnet transmits credentials in plaintext and is considered insecure for production environments.

The Jump Server was used as a secure gateway for administrative access to the OT zone. Engineering users must SSH into the Jump Server first, and from there connect to OT devices. They cannot directly reach OT from their workstations.

During implementation, it was discovered that Cisco Packet Tracer servers cannot originate SSH sessions. As a result, the Jump Server was replaced with a Cisco router configured with SSH, which provides full SSH client and server capability.

Jump Server SSH Credentials:

- Username: admin
- Password: Cisco123

```
Jump-Server(config)#line vty 0 4
Jump-Server(config-line)#login local
Jump-Server(config-line)#transport input ssh
Jump-Server(config-line)#exit
Jump-Server(config)#ip ssh version 2
Jump-Server(config)#no ip routing
Jump-Server(config)#
```

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Figure 10: SSH Jump Server Configuration

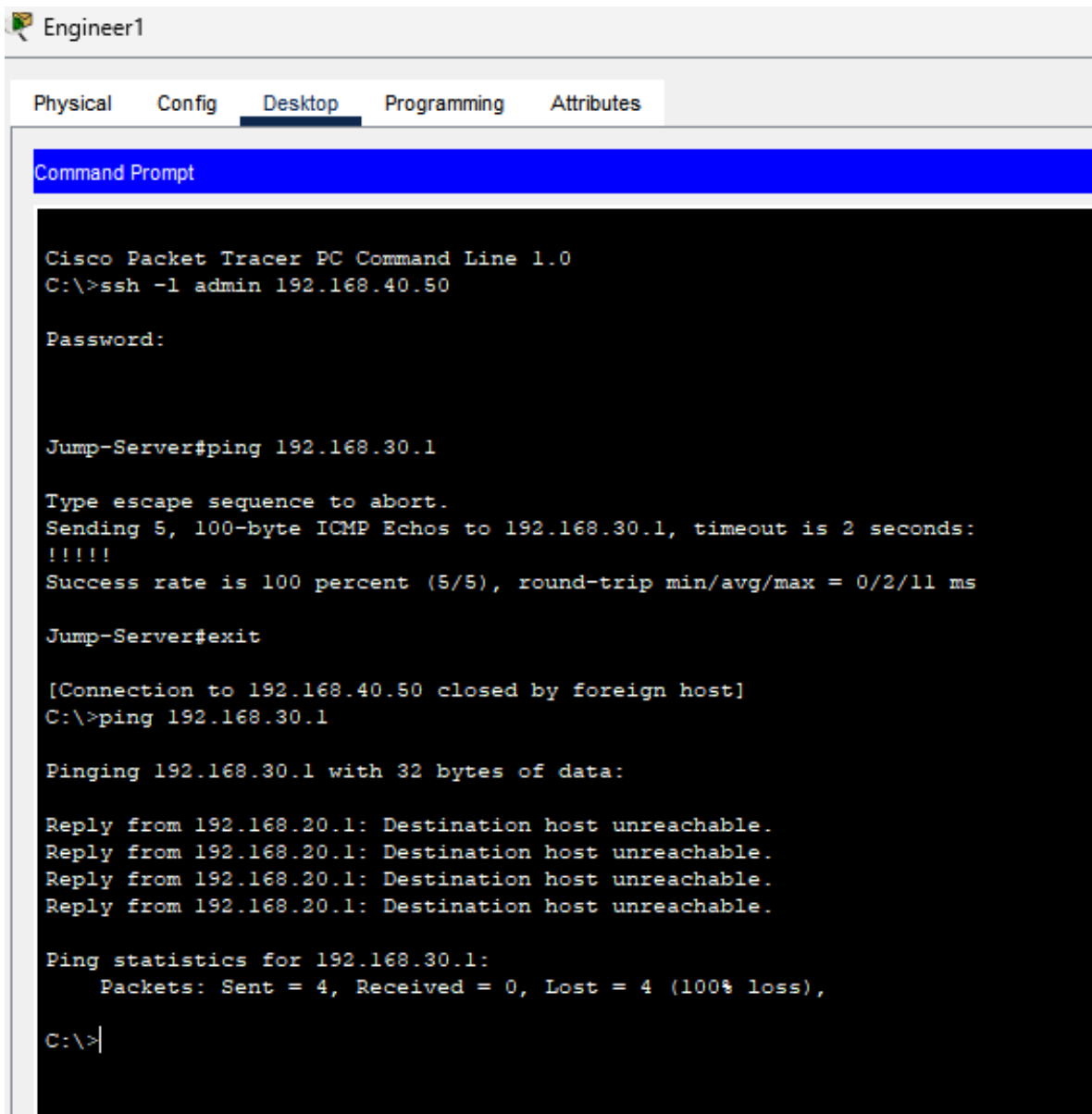


Figure 11: Jump Server SSH Testing

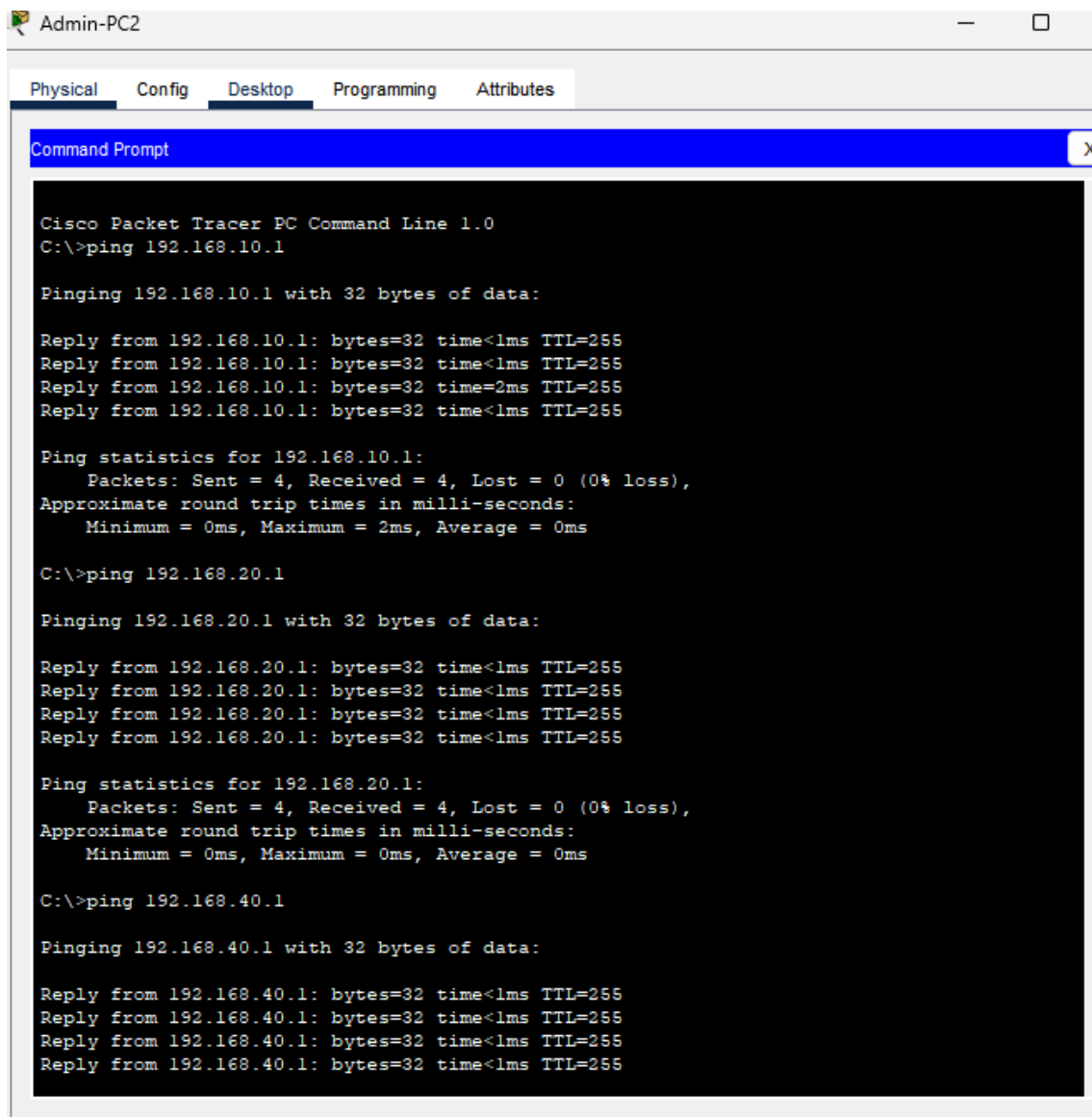
9. Connectivity and Security Testing

The comprehensive testing was conducted to verify both authorized connectivity and security restrictions across all zones. Each test confirms a specific policy or access rule.

9.1 Inter-VLAN Routing Tests

Test performed from IT zone (HR-PC0, 192.168.10.x):

- ping 192.168.10.1 (IT gateway) → SUCCESS
- ping 192.168.20.1 (ENG gateway) → SUCCESS
- ping 192.168.40.1 (DMZ gateway) → SUCCESS
- ping 192.168.30.1 (OT gateway) → BLOCKED (ACL enforced)



```
Admin-PC2
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=2ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
```

Figure 12: Inter-VLAN Routing Test from IT Zone

```
C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 13: Inter-VLAN Routing Test — Additional Verification

9.2 OT Air-Gap Test

Test performed from OT zone devices to verify isolation:

- ping 192.168.10.1 (IT zone) from OT → BLOCKED
- ping 192.168.20.1 (ENG zone) from OT → BLOCKED
- ping 192.168.40.10 (Jump Server) from OT → verified via Jump Server path

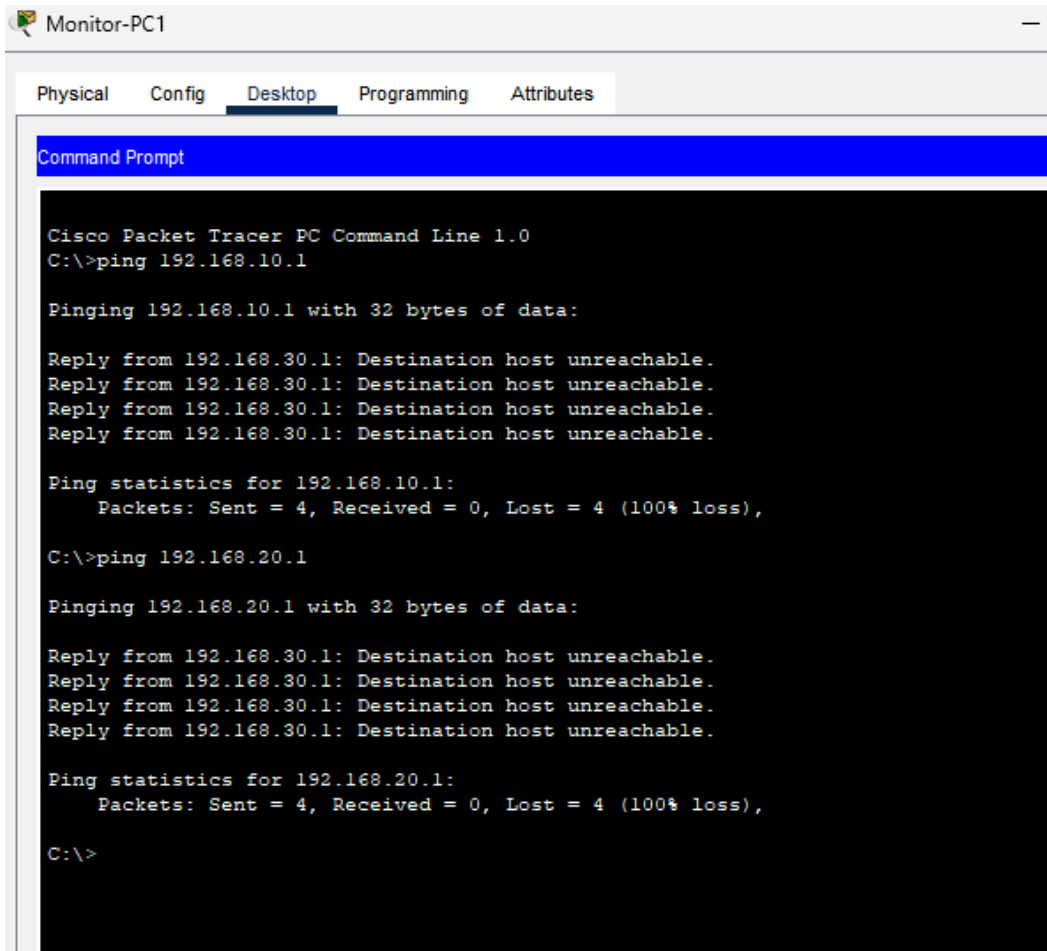


Figure 14: OT Air-Gap Test — IT and ENG Access Blocked from OT

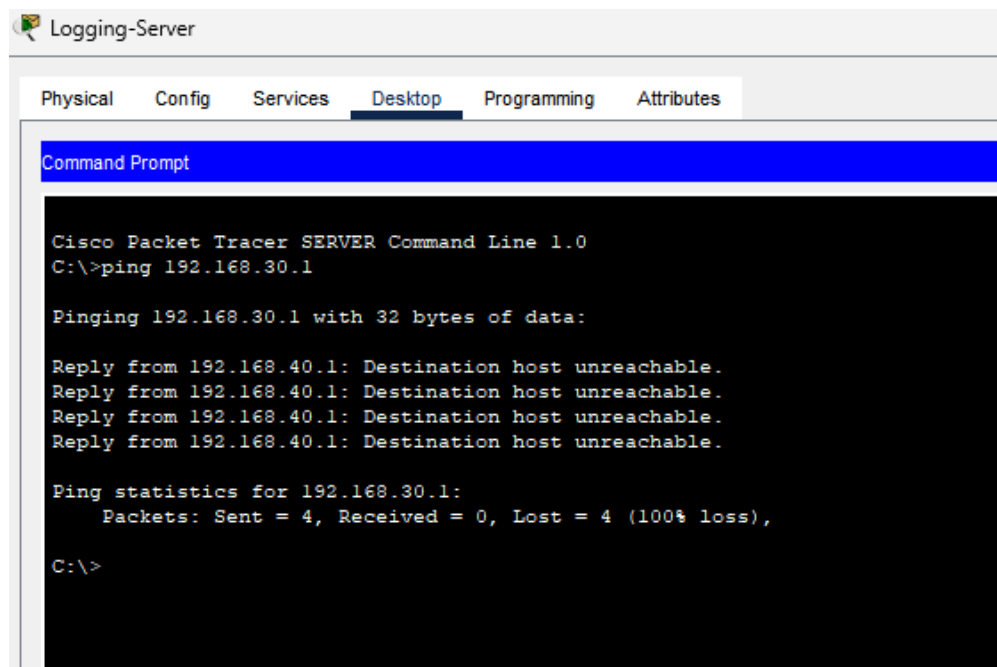
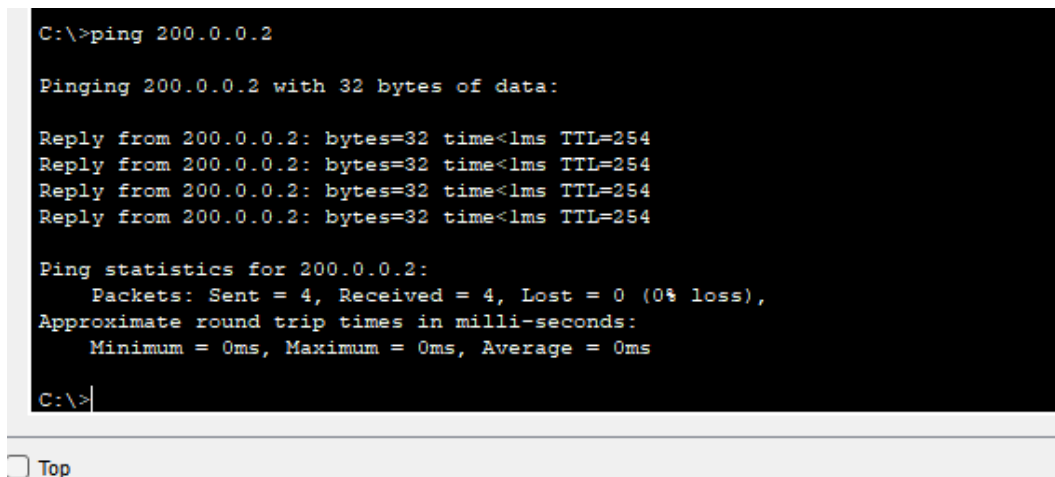


Figure 15: Pinging OT from Logging Server (DMZ)

9.3 Internet Access Restriction Test

Test performed to confirm OT has no internet access and IT has controlled access:

- From IT PC to router internet-side interface → ALLOWED (controlled via DMZ)
- From OT to router internet-side interface → BLOCKED (ACL enforced)
- From OT to router inside interface → BLOCKED



```
C:\>ping 200.0.0.2

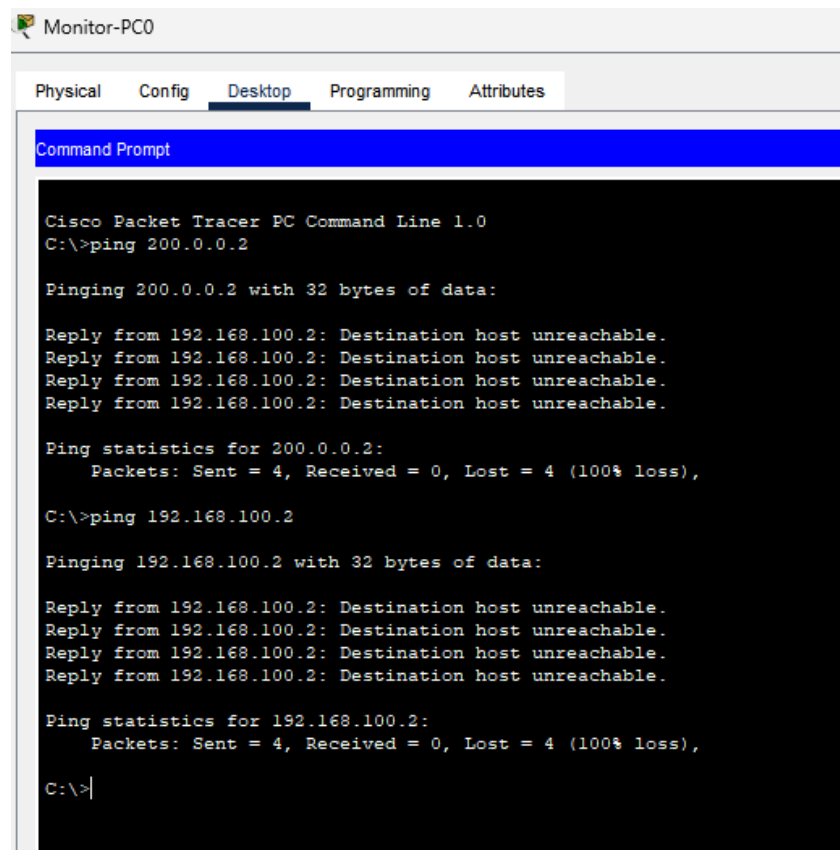
Pinging 200.0.0.2 with 32 bytes of data:

Reply from 200.0.0.2: bytes=32 time<1ms TTL=254
Reply from 200.0.0.2: bytes=32 time<1ms TTL=254
Reply from 200.0.0.2: bytes=32 time<1ms TTL=254
Reply from 200.0.0.2: bytes=32 time<1ms TTL=254

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 16: IT to Router Internet Side — Allowed



```
Monitor-PC0
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:

Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 17: OT to Router Internet Side — Blocked (Policy Enforced)

9.4 Layer 2 Security Verification

The following Layer 2 security features were configured on all access switches to protect against common Layer 2 attacks:

Feature	Applied On
Port Security	All access switches
DHCP Snooping	All access switches
BPDU Guard	All access switches
Dynamic ARP Inspection (DAI)	All access switches

Table 3: Layer 2 Security Features

```
SW-IT(config-if-range)#spanning-tree bpduguard enable
SW-IT(config-if-range)#exit
SW-IT(config)#interface range fa0/1 - 3
SW-IT(config-if-range)#switchport port-security
SW-IT(config-if-range)#switchport port-security maximum 2
SW-IT(config-if-range)#switchport port-security violation restrict
SW-IT(config-if-range)#switchport port-security aging time 10
SW-IT(config-if-range)#exit
SW-IT(config)#interface range fa0/5 - 10
SW-IT(config-if-range)#switchport port-security
SW-IT(config-if-range)#switchport port-security maximum 2
SW-IT(config-if-range)#switchport port-security violation restrict
SW-IT(config-if-range)#switchport port-security aging time 10
SW-IT(config-if-range)#exit
SW-IT(config)#
```

Copy

Figure 18: Layer 2 Security Configuration Verification

SW-DMZ

Physical Config CLI Attributes

IOS Command Line Interface

```
SW-DMZ#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
(Count) (Count) (Count)
-----
Fa0/1 2 0 0 Restrict
Fa0/2 2 0 0 Restrict
Fa0/3 2 1 0 Restrict
Fa0/5 2 0 0 Restrict
Fa0/6 2 0 0 Restrict
Fa0/7 2 0 0 Restrict
Fa0/8 2 0 0 Restrict
Fa0/9 2 0 0 Restrict
Fa0/10 2 0 0 Restrict

SW-DMZ#show spanning-tree
VLAN0040
Spanning tree enabled protocol ieee
Root ID Priority 32808
Address 0002.4AA4.D2CC
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32808 (priority 32768 sys-id-ext 40)
Address 0002.4AA4.D2CC
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface Role Sts Cost Prio.Nbr Type
-----
Fa0/1 Desg FWD 19 128.1 P2p
Fa0/4 Desg FWD 19 128.4 P2p
Fa0/2 Desg FWD 19 128.2 P2p
Fa0/3 Desg FWD 19 128.3 P2p

VLAN0099
Spanning tree enabled protocol ieee
Root ID Priority 32867
Address 0001.9788.064E
Cost 38
Port 4(FastEthernet0/4)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
```

```
Bridge ID  Priority      32867  (priority 32768 sys-id-ext 99)
           Address      0002.4AA4.D2CC
           Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
           Aging Time   20
```

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/4	Root	FWD	19	128.4	P2p

10. AAA, DHCP, and Centralized Services

10.1 AAA (Authentication, Authorization, Accounting)

Centralized authentication was configured using a TACACS+ AAA server (192.168.10.x in the IT zone). All network devices authenticate through this server, ensuring that every login attempt, configuration change, and command execution is logged.

AAA Policy:

- All network devices require TACACS+ authentication
- Default or blank passwords are strictly forbidden
- All admin accounts must use strong passwords (minimum 8 characters)
- Failed login attempts are logged to the Logging Server

```

CoreSwitch(config)#logging 192.168.40.30
CoreSwitch(config)#end
CoreSwitch#
%SYS-5-CONFIG_I: Configured from console by console
%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.40.30 port 514 started - CLI initiated

CoreSwitch#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 36 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 36 messages logged, xml disabled,
                  filtering disabled
Buffer logging: disabled, xml disabled,
                  filtering disabled

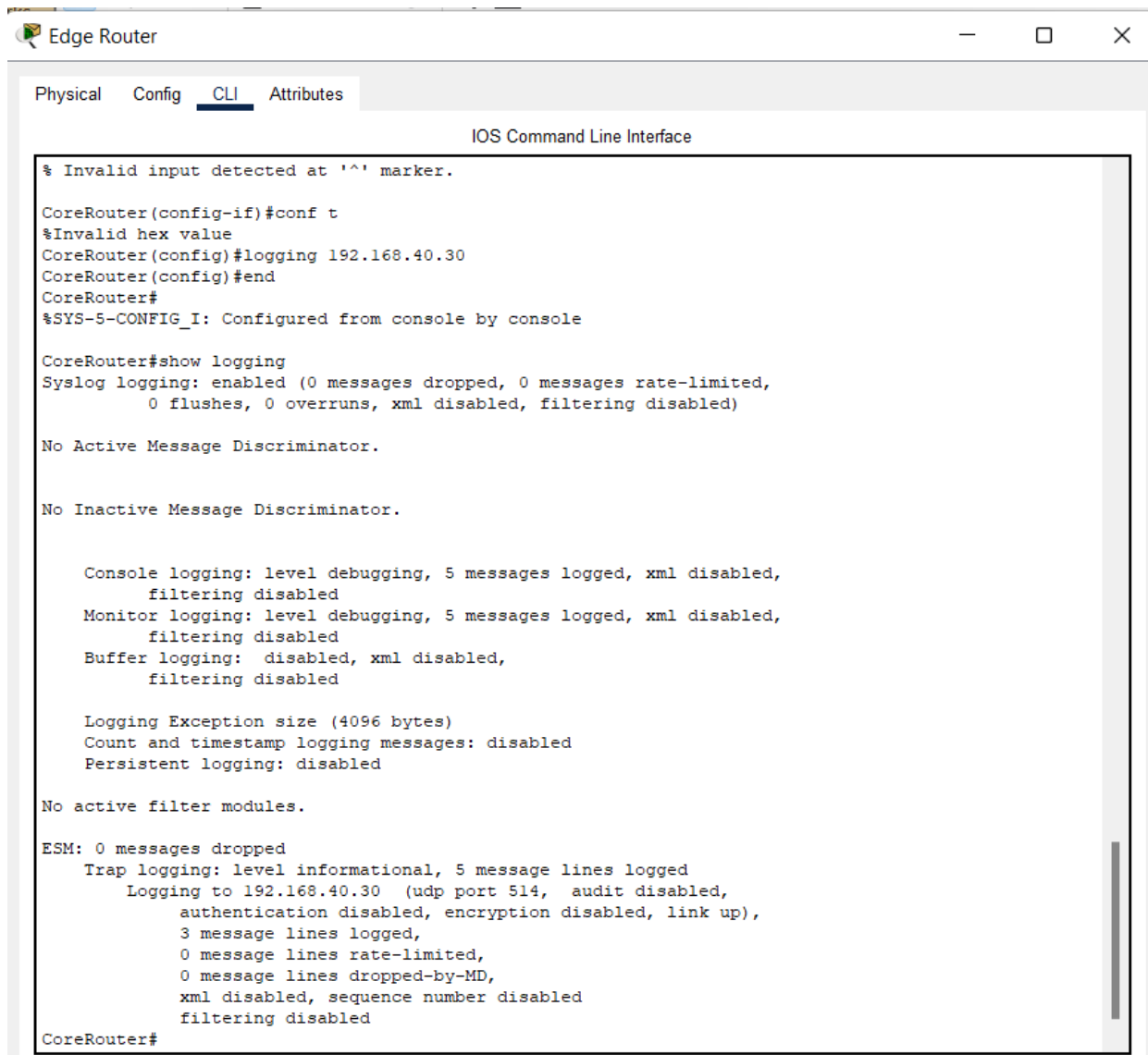
Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 36 message lines logged
  Logging to 192.168.40.30 (udp port 514, audit disabled,
    authentication disabled, encryption disabled, link up),
    2 message lines logged,
    0 message lines rate-limited,
    0 message lines dropped-by-MD,
    xml disabled, sequence number disabled
    filtering disabled
CoreSwitch#

```

Figure 19: AAA Server Configuration

The image shows a web-based interface for an 'Edge Router'. At the top, there are tabs for 'Physical', 'Config', 'CLI', and 'Attributes', with 'CLI' being the active tab. Below the tabs, the title 'IOS Command Line Interface' is displayed. The main area contains a text box with the following text:

```
% Invalid input detected at '^' marker.

CoreRouter(config-if)#conf t
%Invalid hex value
CoreRouter(config)#logging 192.168.40.30
CoreRouter(config)#end
CoreRouter#
%SYS-5-CONFIG_I: Configured from console by console

CoreRouter#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 5 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 5 messages logged, xml disabled,
                  filtering disabled
Buffer logging: disabled, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 5 message lines logged
Logging to 192.168.40.30 (udp port 514, audit disabled,
authentication disabled, encryption disabled, link up),
3 message lines logged,
0 message lines rate-limited,
0 message lines dropped-by-MD,
xml disabled, sequence number disabled
filtering disabled

CoreRouter#
```

Figure 20: Logging Server Configuration on Edge Router

Authentication was verified by logging into the Edge Router via SSH from an IT zone workstation. The login request was forwarded to the AAA server at 192.168.10.22, authenticated successfully, and the session was logged to the Logging Server.

Fallback to local authentication is configured in case the AAA server becomes unreachable, ensuring administrative access is never completely lost.

10.2 DHCP Configuration

DHCP is enabled exclusively for the IT Zone (Zone A). All other zones use static IP addressing to prevent unauthorized devices from obtaining network addresses.

- DHCP Server static IP: 192.168.10.21
- DHCP Snooping is enabled on access switches to prevent rogue DHCP servers
- Engineering, OT, and DMZ zones use fully static IP assignments

The following figures show the steps used for the DHCP configurations:

The screenshot shows the DHCP-Server configuration window. The 'Services' tab is selected, and the 'DHCP' service is configured for the 'FastEthernet0' interface. The service is turned 'On'. The configuration includes a pool named 'serverPool' with a default gateway of 192.168.10.1, a DNS server of 0.0.0.0, and a start IP address of 192.168.10.10 with a subnet mask of 255.255.255.0. The maximum number of users is set to 246. The TFTP server and WLC address are both set to 0.0.0.0. A table at the bottom lists the configured pool.

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.10.1	0.0.0.0	192.168.10.10	255.255.255.0	246	0.0.0.0	0.0.0.0

Figure 21: DHCP Configuration

```
cisCoreSwitch>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	unset	up	up
FastEthernet0/2	unassigned	YES	unset	up	up
FastEthernet0/3	unassigned	YES	unset	up	up
FastEthernet0/4	unassigned	YES	unset	up	up
FastEthernet0/5	192.168.100.1	YES	manual	up	up
FastEthernet0/6	unassigned	YES	unset	down	down
FastEthernet0/7	unassigned	YES	unset	down	down
FastEthernet0/8	unassigned	YES	unset	down	down
FastEthernet0/9	unassigned	YES	unset	down	down
FastEthernet0/10	unassigned	YES	unset	down	down
FastEthernet0/11	unassigned	YES	unset	down	down
FastEthernet0/12	unassigned	YES	unset	down	down
FastEthernet0/13	unassigned	YES	unset	down	down
FastEthernet0/14	unassigned	YES	unset	down	down
FastEthernet0/15	unassigned	YES	unset	down	down
FastEthernet0/16	unassigned	YES	unset	down	down
FastEthernet0/17	unassigned	YES	unset	down	down
FastEthernet0/18	unassigned	YES	unset	down	down
FastEthernet0/19	unassigned	YES	unset	down	down
FastEthernet0/20	unassigned	YES	unset	down	down
FastEthernet0/21	unassigned	YES	unset	down	down
FastEthernet0/22	unassigned	YES	unset	down	down
FastEthernet0/23	unassigned	YES	unset	down	down
FastEthernet0/24	unassigned	YES	unset	down	down
GigabitEthernet0/1	unassigned	YES	unset	down	down
GigabitEthernet0/2	unassigned	YES	unset	down	down
Vlan1	unassigned	YES	unset	administratively down	down
Vlan10	192.168.10.1	YES	manual	up	up

Figure 22: DHCP Configuration check connectivity

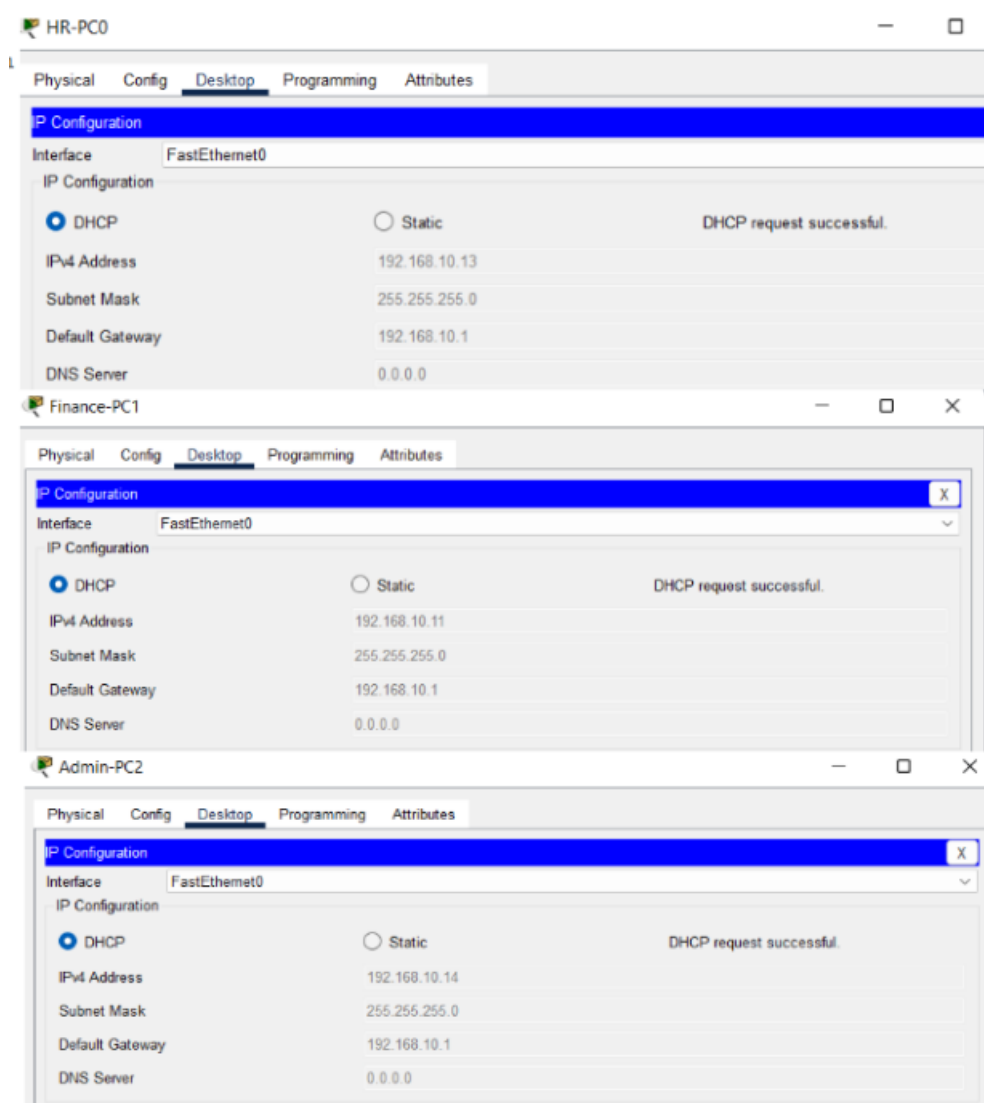


Figure 23: Test DHCP for the PCs in zone A

10.3 Logging Server Configuration

The central Logging-Server (Syslog) in the DMZ was configured to receive logs from all key network devices, providing a single point of audit for security events.

- Logging-Server IP: 192.168.40.30 (static, DMZ zone)
- Syslog service: ON
- Core Switch: 'logging 192.168.40.30': confirmed with LOGGINGHOST_STARTSTOP message

- Edge Router: 'logging 192.168.40.30': confirmed active

```
CoreSwitch(config)#logging 192.168.40.30
CoreSwitch(config)#end
CoreSwitch#
%SYS-5-CONFIG_I: Configured from console by console
%SYS-6-LOGGINGHOST_STARTSTOP: Logging to host 192.168.40.30 port 514 started - CLI initiated

CoreSwitch#show logging
Syslog logging: enabled (0 messages dropped, 0 messages rate-limited,
                  0 flushes, 0 overruns, xml disabled, filtering disabled)

No Active Message Discriminator.

No Inactive Message Discriminator.

Console logging: level debugging, 36 messages logged, xml disabled,
                  filtering disabled
Monitor logging: level debugging, 36 messages logged, xml disabled,
                  filtering disabled
Buffer logging: disabled, xml disabled,
                  filtering disabled

Logging Exception size (4096 bytes)
Count and timestamp logging messages: disabled
Persistent logging: disabled

No active filter modules.

ESM: 0 messages dropped
Trap logging: level informational, 36 message lines logged
  Logging to 192.168.40.30 (udp port 514, audit disabled,
    authentication disabled, encryption disabled, link up),
    2 message lines logged,
    0 message lines rate-limited,
    0 message lines dropped-by-MD,
    xml disabled, sequence number disabled
    filtering disabled
CoreSwitch#
```

Figure 24: Logging Server Configuration for Core Switch

10.4 Wireless Access Point

A single authorized Wi-Fi Access Point is deployed in the IT Zone (Zone A). The SSID is "IT-WiFi" configured with WPA2-PSK encryption. Unauthorized APs connected to any switch port constitute a policy violation.

The steps are:

- Access Point connected to SW-IT FastEthernet0/7
- SW-IT Fa0/7 assigned to VLAN 10 (Access mode)
- Security: WPA2-PSK | Encryption: AES | Password: cisco123
- Wireless laptop (WPC300N card) connected to IT-WiFi successfully
- Connectivity verified via ping to gateway 192.168.10.1

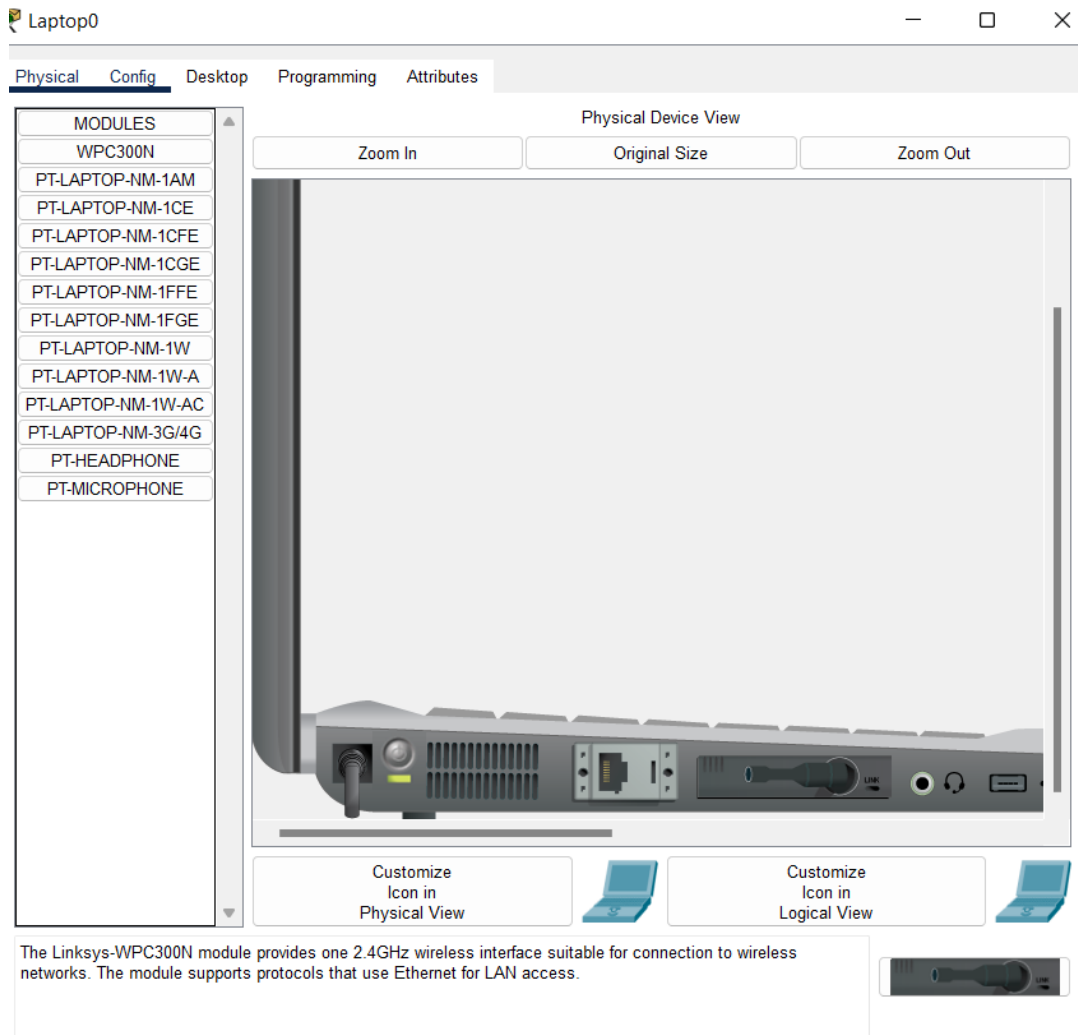


Figure 25: Wi-Fi Access Point physical Configuration

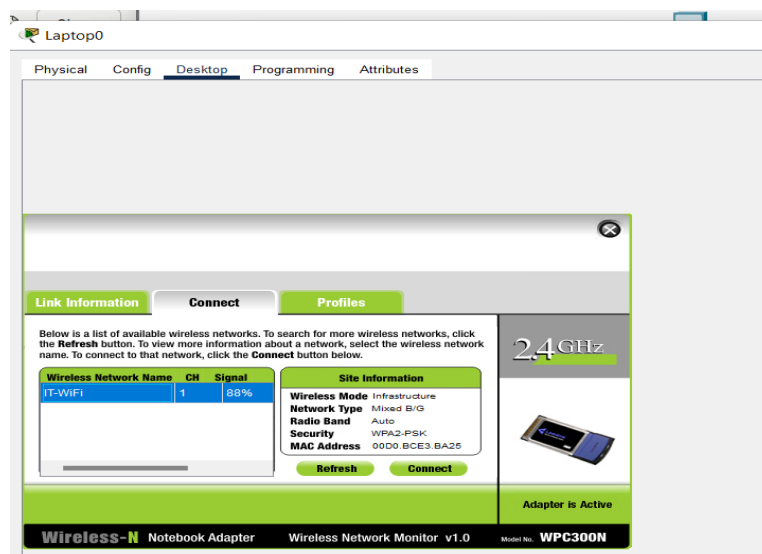


Figure 26: Laptop detecting IT-WiFi network (WPA2-PSK, signal 88%, Adapter Active)

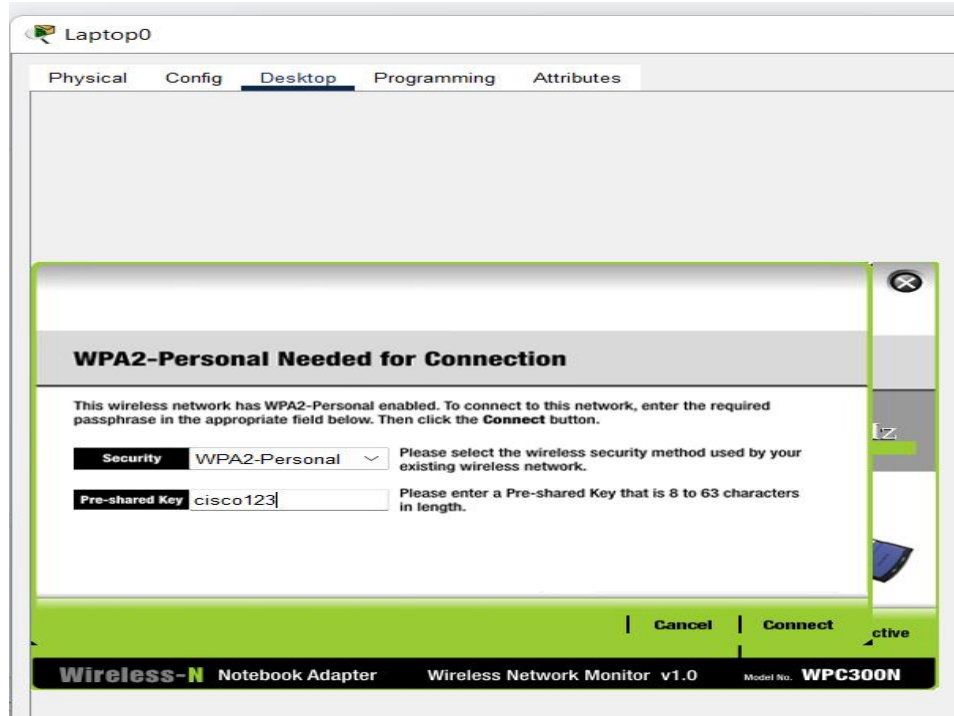


Figure 27: WPA2-Personal authentication with pre-shared key cisco123

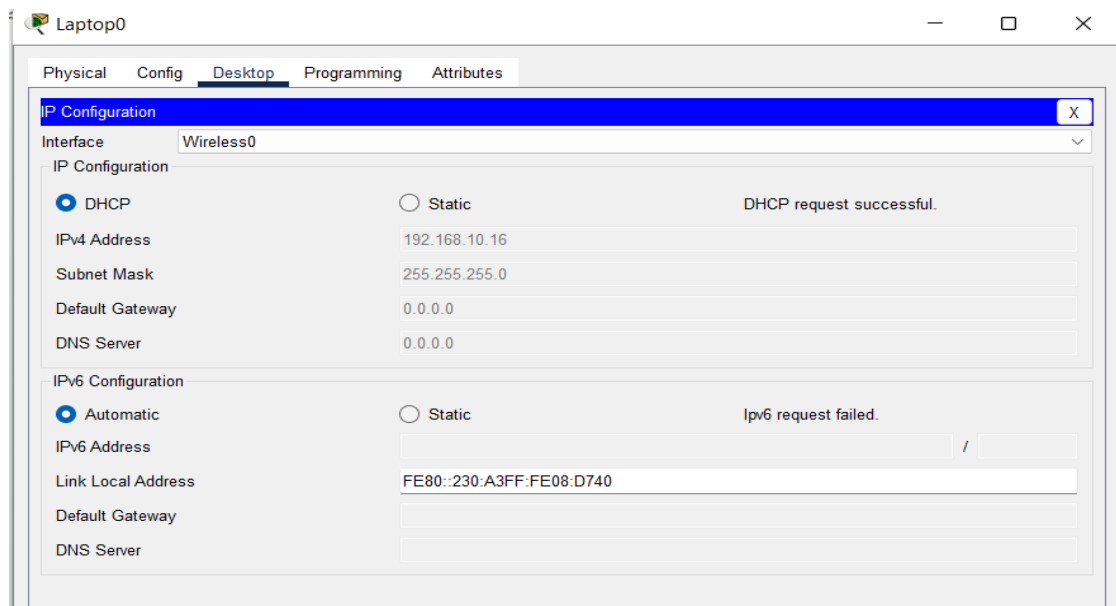


Figure 28: Laptop0 DHCP request successful

10.5 IPsec Site-to-Site VPN

A site-to-site IPsec VPN was configured between the Edge Router and the Remote Router to provide encrypted connectivity for the remote site (192.168.50.0/24). The VPN ensures that remote site traffic is encrypted and cannot be intercepted in transit.

- VPN type: IPsec Site-to-Site
- Peer: Remote Router
- Protected traffic: 192.168.50.0/24 → internal zones
- Direct remote access without VPN is strictly forbidden

```
CoreRouter#show crypto map
Crypto Map VPN-MAP 10 ipsec-isakmp
  Peer = 200.0.0.1
  Extended IP access list 110
    access-list 110 permit ip 192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255
  Current peer: 200.0.0.1
  Security association lifetime: 4608000 kilobytes/3600 seconds
  PFS (Y/N): N
  Transform sets={
    VPN-SET,
  }
  Interfaces using crypto map VPN-MAP:
    GigabitEthernet0/0

CoreRouter#
```

Copy

Paste

Figure 29: IPsec Edge Router VPN Configuration

```
Username: admin
Password:
CoreRouter>enable
Password:
CoreRouter#license boot module c2900 technology-package securityk9
^
% Invalid input detected at '^' marker.

CoreRouter#conf t
Enter configuration commands, one per line. End with CNTL/Z.
CoreRouter(config)#license boot module c2900 technology-package securityk9
PLEASE READ THE FOLLOWING TERMS CAREFULLY. INSTALLING THE LICENSE OR
LICENSE KEY PROVIDED FOR ANY CISCO PRODUCT FEATURE OR USING SUCH
PRODUCT FEATURE CONSTITUTES YOUR FULL ACCEPTANCE OF THE FOLLOWING
TERMS. YOU MUST NOT PROCEED FURTHER IF YOU ARE NOT WILLING TO BE BOUND
BY ALL THE TERMS SET FORTH HEREIN.

Use of this product feature requires an additional license from Cisco,
together with an additional payment. You may use this product feature
on an evaluation basis, without payment to Cisco, for 60 days. Your use
of the product, including during the 60 day evaluation period, is
subject to the Cisco end user license agreement
http://www.cisco.com/en/US/docs/general/warranty/English/EULKEN.html
If you use the product feature beyond the 60 day evaluation period, you
must submit the appropriate payment to Cisco for the license. After the
60 day evaluation period, your use of the product feature will be
governed solely by the Cisco end user license agreement (link above),
together with any supplements relating to such product feature. The
above applies even if the evaluation license is not automatically
terminated and you do not receive any notice of the expiration of the
evaluation period. It is your responsibility to determine when the
evaluation period is complete and you are required to make payment to
Cisco for your use of the product feature beyond the evaluation period.

Your acceptance of this agreement for the software features on one
product shall be deemed your acceptance with respect to all such
software on all Cisco products you purchase which includes the same
software. (The foregoing notwithstanding, you must purchase a license
for each software feature you use past the 60 days evaluation period,
so that if you enable a software feature on 1000 devices, you must
purchase 1000 licenses for use past the 60 day evaluation period.)

Activation of the software command line interface will be evidence of
your acceptance of this agreement.

ACCEPT? [yes/no]: yes
% use 'write' command to make license boot config take effect on next boot

CoreRouter(config)#end
```

Figure 30: Security license activation on router


```

CoreRouter(config)#crypto isakmp policy 10
CoreRouter(config-isakmp)#encryption aes
CoreRouter(config-isakmp)#hash sha
CoreRouter(config-isakmp)#authentication pre-share
CoreRouter(config-isakmp)#group 2
CoreRouter(config-isakmp)#lifetime 86400
CoreRouter(config-isakmp)#exit
CoreRouter(config)#crypto isakmp policy 10
CoreRouter(config-isakmp)# encryption aes
CoreRouter(config-isakmp)# hash sha
CoreRouter(config-isakmp)# authentication pre-share
CoreRouter(config-isakmp)# group 2
CoreRouter(config-isakmp)# lifetime 86400
CoreRouter(config-isakmp)#exit
CoreRouter(config)#crypto isakmp key cisco123 address 200.0.0.1
CoreRouter(config)#access-list 110 permit ip 192.168.50.0 0.0.0.255 192.168.10.0
0.0.0.255
CoreRouter(config)#crypto ipsec transform-set VPN-SET esp-aes esp-sha-hmac
CoreRouter(config)#crypto map VPN-MAP 10 ipsec-isakmp
% NOTE: This new crypto map will remain disabled until a peer
and a valid access list have been configured.
CoreRouter(config-crypto-map)# set peer 200.0.0.1
CoreRouter(config-crypto-map)# set transform-set VPN-SET
CoreRouter(config-crypto-map)# match address 110
CoreRouter(config-crypto-map)#exit
CoreRouter(config)#int g0/0
CoreRouter(config-if)# crypto map VPN-MAP
*Jan 3 07:16:26.785: %CRYPTO-6-ISA_KMP_ON_OFF: ISAKMP is ON
CoreRouter(config-if)#end
CoreRouter#
%SYS-5-CONFIG_I: Configured from console by console
write
Building configuration...
[OK]
CoreRouter#show crypto isakmp sa
IPv4 Crypto ISAKMP SA
dst src state conn-id slot status

IPv6 Crypto ISAKMP SA

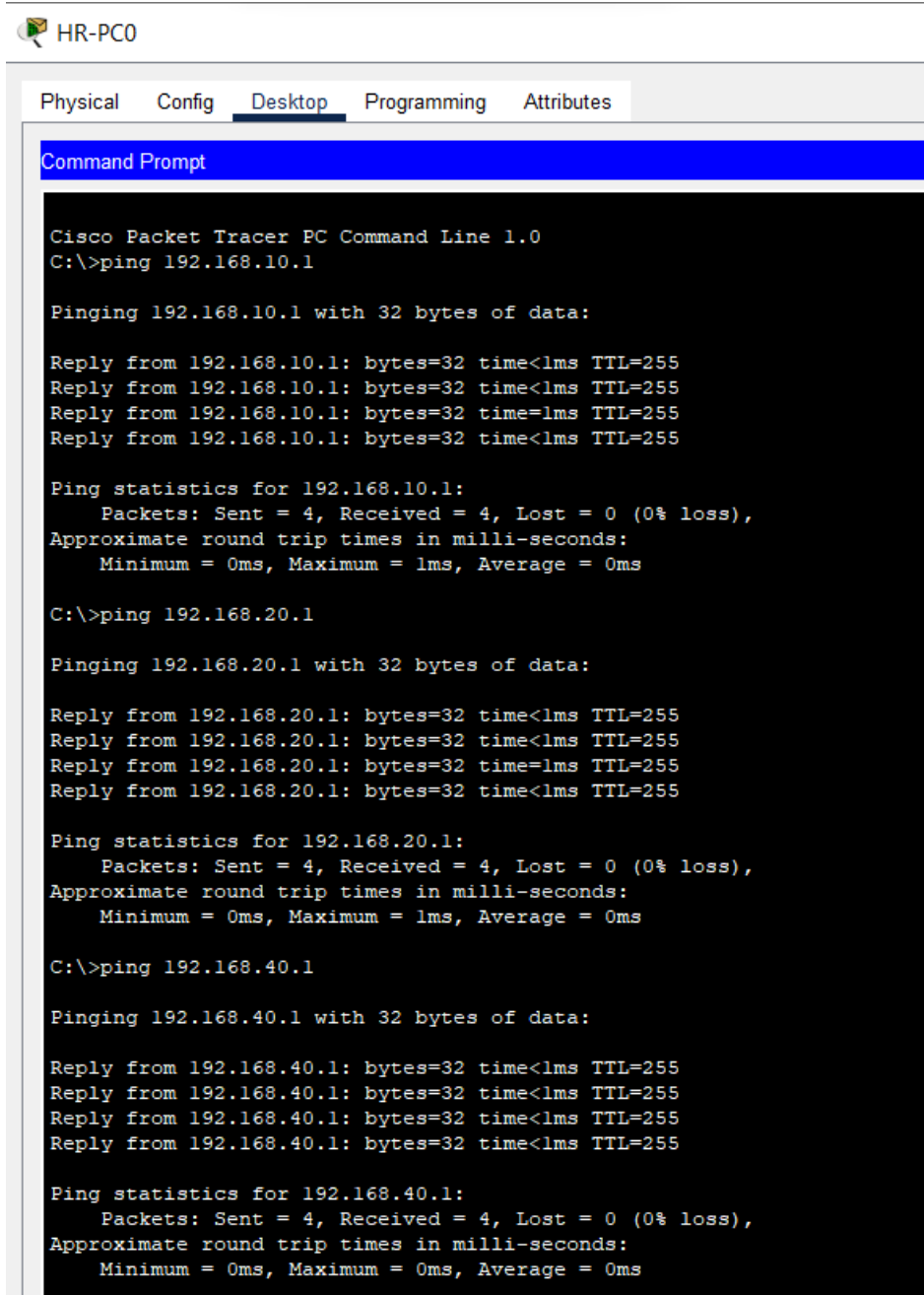
```

Figure 31: IPsec VPN configuration on router

11. Traffic Capture and Monitoring

Wireshark packet captures were used to validate traffic flow, verify that ACL rules are dropping unauthorized packets, and confirm that legitimate traffic passes through correctly.

11.1 HR-PC0 Connectivity Test



The screenshot shows the 'HR-PC0' window with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the results of three ping tests performed from the PC's command line. The tests are for 192.168.10.1, 192.168.20.1, and 192.168.40.1. Each test shows four successful replies with 32 bytes of data, a time of less than 1ms, and a TTL of 255. The statistics for each test indicate that all four packets were sent and received, with 0% loss, and the round trip times were all 0ms.

```
HR-PC0
Physical  Config  Desktop  Programming  Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time=1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Figure 32: HR-PC0 Connectivity Test Results

11.2 Wireshark Captures

```
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Figure 33: Wireshark Capture Overview for HR-PC0

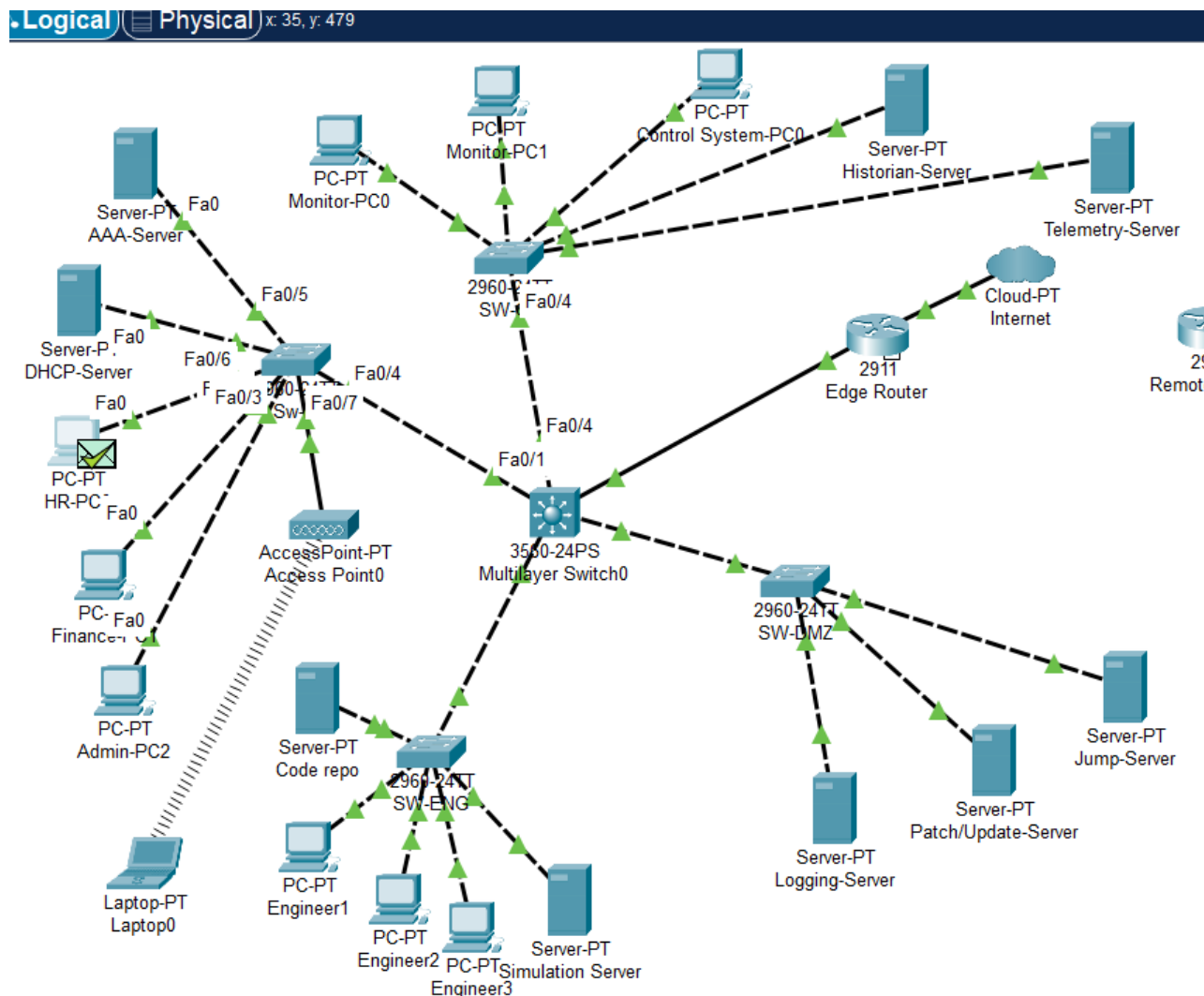


Figure 34: Wireshark Capture for Data Transfer from HR-PC0

PDU Information at Device: Multilayer Switch0

OSI Model

Outbound PDU Details

At Device: Multilayer Switch0
Source: Multilayer Switch0
Destination: HR-PC0

In Layers

Layer7
Layer6
Layer5
Layer4

Layer3

Layer2
Layer1

Out Layers

Layer7
Layer6
Layer5
Layer4
Layer 3: IP Header Src. IP: 192.168.10.1,
Dest. IP: 192.168.10.10 ICMP Message
Type: 3 ICMP Message Type: 8
Layer 2: Dot1q Header 00E0.B045.D501 >>
0000.0C5E.4D00
Layer 1: Port(s): FastEthernet0/1

1. The device sends back an ICMP Administratively Prohibited Unreachable message.
2. The device looks up the destination IP address in the routing table.
3. The routing table finds a routing entry to the destination IP address.
4. The destination network is directly connected. The device sets destination as the next-hop.

Challenge Me

<< Previous Layer

Next Layer >>

Figure 35: Dropped Packet Capture (HR-PC0 to SW-IT)

12. Threat Analysis

The following threats were identified during the design phase and mitigated through the implemented security controls:

#	Threat	Zone Affected	Description
1	Unauthorized OT Access	OT Zone	Attackers try to directly access OT devices, bypassing the Jump Server
2	DHCP Spoofing	IT Zone (Zone A)	A rogue DHCP server gives fake IPs/gateways to IT zone PCs, redirecting traffic
3	Syslog Tampering	DMZ	Attackers target the Logging-Server to delete or modify logs, hiding malicious activity
4	Brute Force on AAA	All Zones	Repeated login attempts on network devices to guess credentials and gain admin access
5	Rogue Wireless AP	IT Zone	Unauthorized AP connected to SW-IT, allowing unauthorized wireless access
6	VPN Credential Theft	Remote Site	VPN credentials are stolen, allowing an attacker to connect from the remote site
7	Lateral Movement ENG→OT	ENG/OT Zone	Compromised engineer workstation used to pivot into the OT zone
8	OT Internet Access	OT Zone	OT devices attempt to communicate with external internet, exposing critical systems
9	Unencrypted Traffic Interception	All Zones	Sensitive data between zones captured via man-in-the-middle attack
10	Jump Server Compromise	DMZ	If Jump Server is compromised, attackers gain direct SSH access to all OT devices

Table 4: Threat Analysis

13. Risk Assessment

#	Threat	Likelihood	Impact	Risk Level
1	Unauthorized OT Access	Medium	Critical	HIGH
2	DHCP Spoofing	Medium	High	HIGH
3	Syslog Tampering	Low	High	MEDIUM
4	Brute Force on AAA	High	Critical	HIGH
5	Rogue Wireless AP	Medium	High	HIGH
6	VPN Credential Theft	Low	Critical	MEDIUM
7	Lateral Movement ENG→OT	Medium	Critical	HIGH
8	OT Internet Access	Low	Critical	MEDIUM
9	Unencrypted Traffic Interception	Medium	High	HIGH
10	Jump Server Compromise	Low	Critical	MEDIUM

Table 5: Risk Assessment Matrix

Risk Level Key:

- *HIGH* for Immediate action required
- *MEDIUM* for Monitor and mitigate
- *LOW* for Acceptable with controls in place

14. Security Policies

Policy 1: Zone Isolation

- IT zone (Zone A) cannot directly communicate with OT zone (Zone C)
- All OT access must go through the Jump Server via SSH only
- ENG zone (Zone B) has limited access to OT — no direct unrestricted communication

Policy 2: Internet Access

- OT zone devices have no internet access under any circumstance
- Internet access is only permitted through the Edge Router for authorized zones
- All internet-bound traffic passes through the DMZ

Policy 3: Authentication (AAA)

- All network devices require TACACS+ authentication via AAA-Server
- Default/blank passwords are strictly forbidden
- All admin accounts must use strong passwords (minimum 8 characters)
- Failed login attempts are logged to the Logging-Server

Policy 4: DHCP

- DHCP is enabled only for IT Zone (Zone A)
- All other zones use static IP addressing
- DHCP-Server is assigned a static IP (192.168.10.21) and monitored for rogue server activity

Policy 5: Logging and Monitoring

- All network devices must send logs to the central Logging-Server (192.168.40.30) in the DMZ
- Logs must be retained and reviewed regularly
- Any device not sending logs is considered a security incident

Policy 6: Wireless Access

- Only one authorized AP is permitted in IT Zone (Zone A)
- SSID: IT-WiFi with WPA2-PSK encryption minimum
- Unauthorized APs connected to any switch port are a policy violation

Policy 7: VPN

- Remote site connectivity is only allowed via IPsec Site-to-Site VPN
- Direct remote access without VPN is strictly forbidden
- VPN credentials must be rotated regularly

Policy 8: DMZ Access

- DMZ servers (Logging, Jump, Patch) are accessible from internal zones but not directly from the internet
- The Jump Server is the only path to reach OT zone devices
- Patch/Update Server is used exclusively for authorized system updates

15. Conclusion

This project successfully designed and implemented a segmented, multi-zone ICS network in Cisco Packet Tracer, applying core cybersecurity principles including Defence in Depth, Zero Trust, and Least Privilege.

The key security goals achieved include:

- Full VLAN segmentation separating IT, Engineering, OT, and DMZ environments
- Inter-VLAN routing controlled by ACLs ensuring no unauthorized cross-zone traffic
- OT zone completely isolated from the internet and unreachable from IT/ENG directly
- Jump Server as the single access path to OT, providing auditable SSH-based management
- Centralized AAA authentication and logging via TACACS+ and the DMZ Logging Server
- Wi-Fi access point secured with WPA2-PSK in the IT zone only
- Site-to-site IPsec VPN securing remote site connectivity
- Layer 2 security (port security, DHCP snooping, BPDU Guard, DAI) protecting access switches

The design demonstrates that even in a simulated environment, rigorous zone separation and enforcement of security policies can significantly reduce the attack surface of an industrial control system network. The identified threats in the threat analysis are mitigated by the layered defenses implemented throughout the infrastructure.

16. Successful testing snippets

Working Demo

(<https://drive.google.com/drive/folders/14N2T0cv0d3z3Ox14SgT3hg8EhvZ87Tb5?usp=sharing>)

Inter-VLAN Routing:

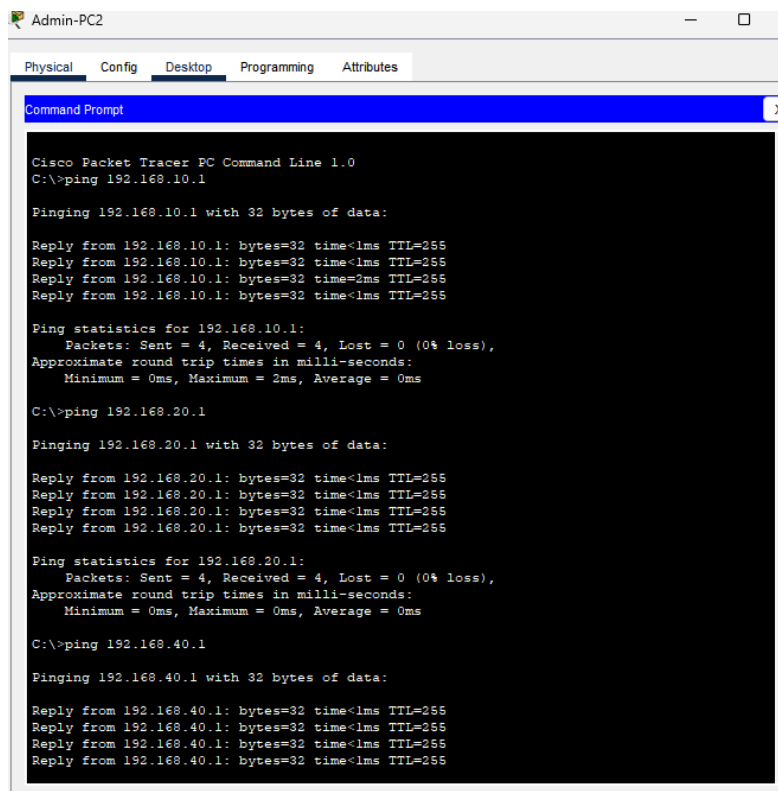
From IT:

ping 192.168.10.1 → IT gateway -allowed

ping 192.168.20.1 → ENG gateway -allowed

ping 192.168.40.1 → DMZ gateway -allowed

ping 192.168.30.1 → OT gateway - blocked



```
Admin-PC2
Physical Config Desktop Programming Attributes
Command Prompt X
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time=2ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255
Reply from 192.168.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
```



```

C:\>ping 192.168.40.1

Pinging 192.168.40.1 with 32 bytes of data:

Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255
Reply from 192.168.40.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.40.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.
Reply from 192.168.10.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

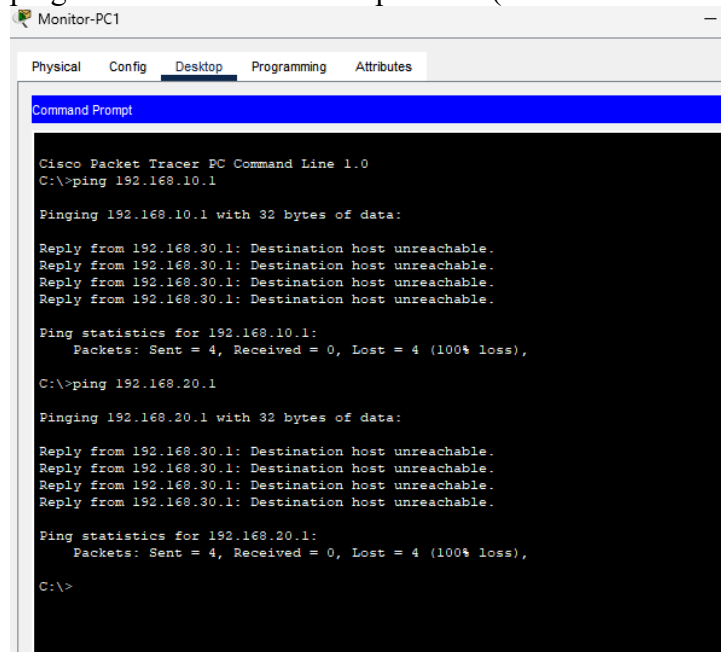
```

OT air-gapped Testing:

ping 192.168.10.1 → IT blocked

ping 192.168.20.1 → ENG blocked

ping 192.168.40.10 → Jump Server (Still needed after replacement)



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'Monitor-PC1'. The window has tabs for Physical, Config, Desktop, Programming, and Attributes, with 'Desktop' selected. The Command Prompt shows the following output:

```

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.20.1

Pinging 192.168.20.1 with 32 bytes of data:

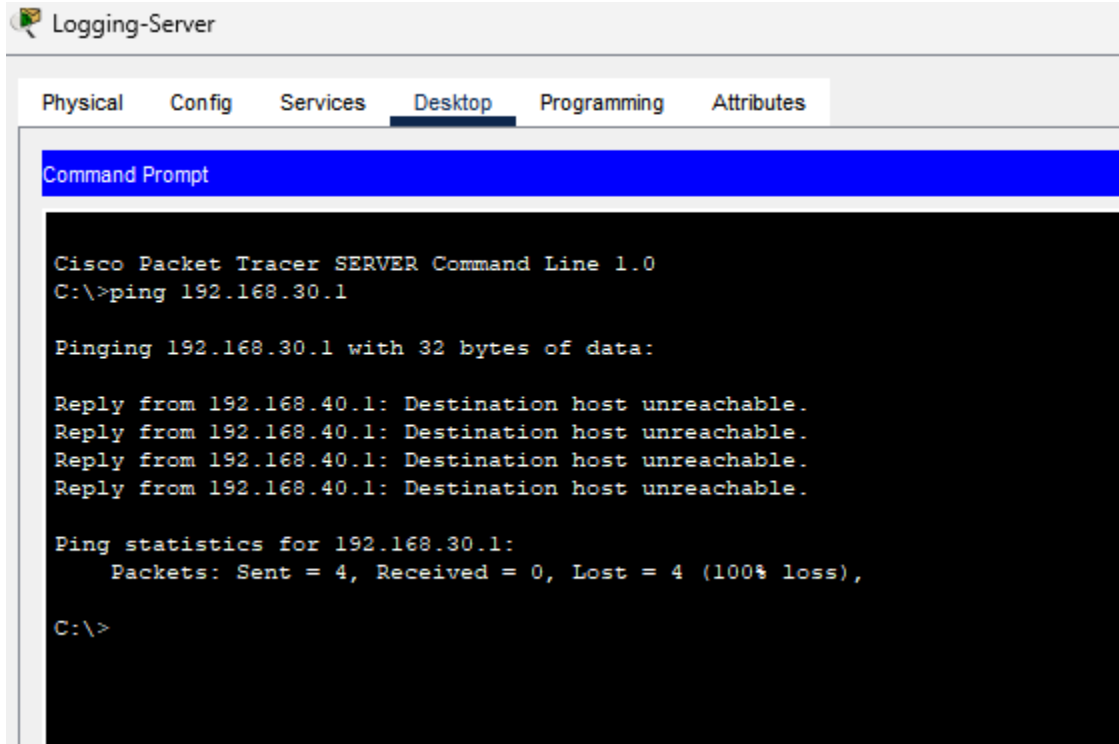
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.
Reply from 192.168.30.1: Destination host unreachable.

Ping statistics for 192.168.20.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>

```

Pinging OT from Logging -DMZ server



```
Logging-Server
Physical  Config  Services  Desktop  Programming  Attributes

Command Prompt

Cisco Packet Tracer SERVER Command Line 1.0
C:\>ping 192.168.30.1

Pinging 192.168.30.1 with 32 bytes of data:

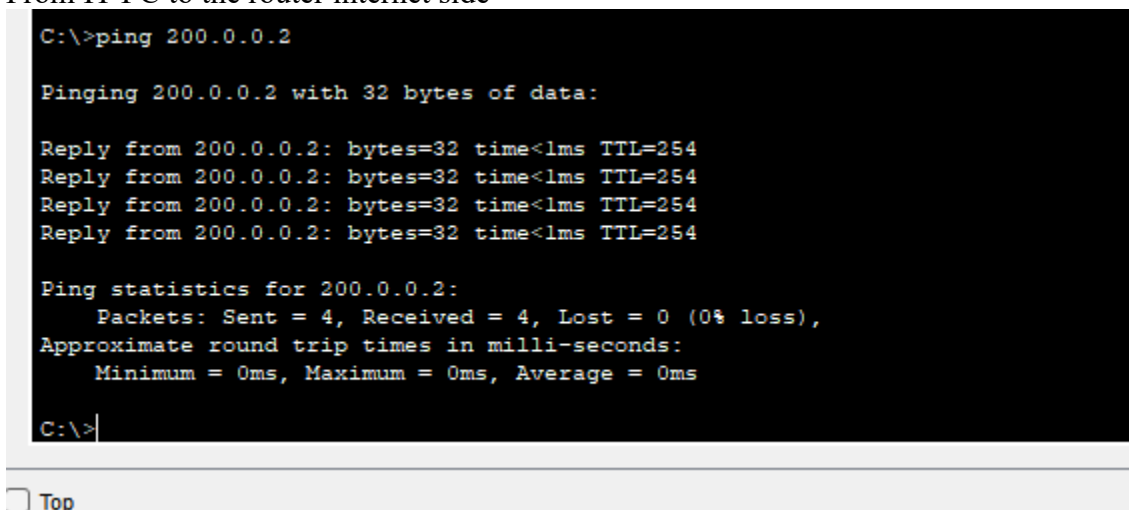
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.
Reply from 192.168.40.1: Destination host unreachable.

Ping statistics for 192.168.30.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Test the router firewall:

From IT PC to the router internet side



```
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 200.0.0.2: bytes=32 time<lms TTL=254
Reply from 200.0.0.2: bytes=32 time<lms TTL=254
Reply from 200.0.0.2: bytes=32 time<lms TTL=254
Reply from 200.0.0.2: bytes=32 time<lms TTL=254

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

From the OT to the router internet side and router inside: Blocked as OT is not allowed to reach the internet:

Monitor-PC0

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:

Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.
Reply from 192.168.100.2: Destination host unreachable.

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>|
```

Testing ZBF (Zone-Based Firewall):

Admin-PC2

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.100.2

Pinging 192.168.100.2 with 32 bytes of data:

Request timed out.
Request timed out.
Reply from 192.168.100.2: bytes=32 time<1ms TTL=254
Reply from 192.168.100.2: bytes=32 time<1ms TTL=254

Ping statistics for 192.168.100.2:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253
Reply from 10.0.0.2: bytes=32 time<1ms TTL=253

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 200.0.0.2

Pinging 200.0.0.2 with 32 bytes of data:

Request timed out.
Reply from 200.0.0.2: bytes=32 time<1ms TTL=253
Reply from 200.0.0.2: bytes=32 time<1ms TTL=253
Reply from 200.0.0.2: bytes=32 time=3ms TTL=253

Ping statistics for 200.0.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>
```

Router ACLs Working:

```
-----  
CoreRouter#sh ip access-lists  
Extended IP access list OUTBOUND  
 10 deny ip 192.168.30.0 0.0.0.255 any (8 match(es))  
 20 deny ip 192.168.31.0 0.0.0.255 any  
 30 permit ip 192.168.10.0 0.0.0.255 any (12 match(es))  
 40 permit ip 192.168.11.0 0.0.0.255 any  
 50 permit ip 192.168.12.0 0.0.0.255 any  
 60 permit ip 192.168.20.0 0.0.0.255 any  
 70 permit ip 192.168.21.0 0.0.0.255 any  
 80 permit ip 192.168.40.0 0.0.0.255 any  
Extended IP access list INBOUND  
 10 permit tcp any any established  
 20 permit icmp any any echo-reply  
 30 permit udp any any eq isakmp  
 40 permit esp any any  
 50 deny ip any any
```